

DATA PROTECTION
FOR 25 YEARS THE COMPASS
OF THE FUTURE

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Acts of the Conference on the
Occasion of the Celebrations for the
25th Anniversary of the
Guarantor for the protection of personal data
Turin, July 4, 2022
edited by
Agostino Ghiglia

DATA PROTECTION
FOR 25 YEARS THE COMPASS
OF THE FUTURE

CONFERENCE ON THE OCCASION OF THE CELEBRATIONS
FOR THE 25TH ANNIVERSARY OF THE GUARANTOR
FOR THE PROTECTION OF PERSONAL DATA

TURIN, JULY 4, 2022

edited by
Agostino Ghiglia

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IT IS POSSIBLE TO VIEW THE RECORDING

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PREFACE

Dr. Agostino Ghiglia

Member of the Guarantor for the protection of personal data

Have 26 years passed or 1 century?

Last year, starting from the Protomoteca hall in Rome,
passing through the Royal Palace of Turin and the Museum of
Pietrarsa in Naples,

concluding with a beautiful meeting with the students at the Teatro

Argentina, we celebrated the first 25 years of the Guarantor for the Protection of Personal Data.

These have been intense years that have not only accompanied the history of Italians but have modified customs, habits, and established practices, often subtly, sometimes with important initiatives from the Authority.

For 25 years, therefore, we have been talking about Privacy or, better, about the protection of personal data, sometimes inappropriately, sometimes (still) with a bored sigh but, thanks to the activity of the Guarantors who have succeeded one another in the last quarter of a century, more and more often with the awareness that the protection of our personal data and our privacy is equivalent to the protection of our freedom.

The “right to be let alone,” that is, the right to protect our most intimate sphere, to the solitude of the self, and the “right to be forgotten,” that is, the right to oblivion and the deletion of our data (and our history) when they are no longer necessary for the purposes for which they were collected

(and their publicity), have become part of our imagination:

we perceive the necessity for such rights to exist.

I would like to take a digression before proceeding with the argument: I am personally and profoundly convinced of the deeply symbolic power of words, of emotional communication.

If, in fact, we were to erase the term privacy and replace it, always, with “personal data protection” and, in Italy, with “protezione dei dati personali,”

I believe we would be much more “popular,” immediately perceived as indispensable (especially in the emerging digital era) and the GDPR

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would become more than a regulation, a valuable “guideline” for the lifestyle of the 21st century. Speaking of “protection” and “personal,” in fact, our amygdala is immediately struck by ancestral calls aimed at preserving our intimacy (and the serenity of our digital twin), thus transforming the huff towards “the umpteenth rule” into a sigh towards “the” rule. For this reason, it is with pride that, in the

wake of the illustrious figures who have spread and fostered awareness of personal data protection, we strive every day to cultivate and refine it through our activities.

Having made this premise, the question I ask myself is: have 25 years passed or a century? The answer is made even more difficult by a technological acceleration to which humanity is not yet prepared and, in most cases, not even aware; a leap that presents a true test of endurance: the protection of the rights that safeguard the dignity of individuals. Improving awareness of data protection in the digital age, indeed, is an extremely arduous task because it means defending and preserving fundamental rights against the private giants of technology, who are more focused on business than on the protection of individuals, but also, sometimes, against public actors who, often in good faith and in an attempt to provide quick responses to citizens, risk losing sight of the indispensable balance between rights and efficiency, between the defense of citizens and invasiveness in their lives, between necessary presence and pervasive immanence in their private sphere.

In this regard, I particularly wish to emphasize the function of preserving the right to personal data protection in the digital age, characterized by “fleeting thought” and the speed imposed by the proposal of ever-new and increasingly rapid and superficially evanescent lifestyles. It is therefore essential to safeguard the protection of personal data, the right to our intimacy, as it is what does not change and must not change, allowing itself to be inertially swept away by the wind of technological progress, that of the digital age, which does not represent a simple evolution but a true social, cultural, economic, and even anthropological revolution.

But beyond this “high” task of guardians of personal data, there are the myriad issues of an increasingly complex daily life that sees the GDPR as an unavoidable hub and bastion of rights in the third millennium: from virtual reality to extensive remote surveillance when not mass surveillance, from the new frontiers of predictive medicine based on data to smart cities, to gatekeepers (“new tyrants” identified in the large international providers, those who manage access to our digital life) of Rifkinian memory with the new blue ticks of Twitter and Meta, from relational chatbots to apps that imitate voice, from revenge porn to cyberbullying, from data breaches to the misdirected camera of the neighbor.... a summary of voices and skills partly ancient and largely new in an era where the personal data circulating through our digital twin becomes increasingly central in our daily lives.

For all these reasons, and for many others that I cannot imagine as we go to print, a stronger and enhanced Data Protection Authority is essential, in its staff, its intelligence, and transformed in its skills, to continue to be a vanguard and at the same time a vigilant guardian in the protection of rights. The most arduous task, finally, that we feel as a duty among duties, is to contribute to educating and protecting new generations, creating in them - the protagonists of a new era - the awareness that technology is two-faced and that they must grow up learning to master it and not to become unaware subjects and passive consumers; and, in this regard, I would also like to reiterate the idea of digital civic education as a subject of study in schools: only the awareness of the new era and its languages can form the individual and collective consciousness of new generations.

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For all the aforementioned reasons, after briefly considering having my thoughts drafted by a chatbot and thinking of having them read with the voice of my favorite football coach, I thought it would be better, in a human-centered view of the law, to do it myself, without forgetting that, I conclude with this reflection, something and not someone could have replaced me.

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Institutional Greetings

- Prof. Stefano Lo Russo
Mayor of the City of Turin

- Hon. Alberto Cirio
President of the Piedmont Region
- Sen. Gilberto Pichetto Fratin
Deputy Minister of Economic Development

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Greetings from the Mayor of the City of Turin
Prof. Stefano Lo Russo

Thank you all, I am very pleased to be here in this place so important for the history of our city. I thank the Authority for choosing our city as the venue for this national conference to celebrate the 25 years of activity of the Authority.

You will also allow me some political reflections on current events: today we celebrate 25 years of an Institution that our country has established as an organ for the protection of citizens, as well expressed by the member Hon. Agostino Ghiglia. Privacy is one of the conditions of a liberal democracy, where the rights protecting the personal sphere have gained increasing interest among citizens and also in the regulations. In a democracy where sovereignty belongs to the people through a democratic mechanism for selecting the ruling class, citizens are guaranteed respect for the prerogatives of their fundamental liberal rights. Never has the issue of guarantees for freedoms been so current, especially in this complex phase of European life, with a war of invasion in Europe, a war that no one imagined and no one could foresee. In this case, we are witnessing the violation of the most evident of the principles underlying international law: the inviolability of borders and the will of the people regarding their sovereignty.

The role of the EU is absolutely growing and strategic when we talk about defense, even through new technologies, thus assuming a central role in the global competition of the Digital Age of the 21st century, and we understand the ongoing resistance from Big Tech to the evolution of European regulations regarding data circulation and the role of privacy in the digital age.

It is evident how today's topic, the activity of the Authority, becomes essential. It was already essential 25 years ago, when we did not have mobile phones in our pockets, when there was not even the Internet, when everything that concerned us had significantly less accessibility. Now, with the digital revolution, these issues have forcefully entered our daily lives.

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Finally, my best wishes for the conduct of the conference. Obviously, my support for your activity also aligns with the perspective, and here I conclude, recalling what the member of the Authority's College Agostino Ghiglia said: it is good that those who have the responsibility of managing sensitive data maintain a high level of autonomy and impartiality. It is a principle that protects everyone. Today, the ownership of data, especially in the digital world, becomes an essential condition, becoming a condition and economic factor. The factor of political influence is also decisive. And there are numerous cases worldwide. Therefore, the work and role of the Authority is crucial, which is not only about formal compliance with the Italian community regulations of guarantee and protection, but it becomes a substantial role in maintaining the democratic balance, in enhancing what I reiterate is the very concept of liberal democracy, and it is, even more so now, at a time when all our lives are essentially constantly interconnected.

So, good work President, good work to the entire College.

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Greetings from the President of the Piedmont Region
Hon. Alberto Cirio

Thank you Agostino and good morning to everyone.

Welcome to Piedmont, welcome to Turin.

First of all, welcome President, welcome to the entire College, a cordial greeting to all the Authorities:

military authorities, civil authorities, and religious authorities who are with us today.

We have opened, Mr. President, the good parlor, as is done in country houses, where you know there was that parlor where one would only pass through but would not sit down unless important guests arrived.

For us, you are important guests, not only because your presence here comes through the precious energy of our Piedmont, which is Agostino Ghiglia, who is a person of value, who today sits within his College, Mr. President, but also especially because we believe very much in your work. And above all in the necessity, as you write in the acts of your conference today, to accompany the Institutions towards what is the future.

This word "accompany" is fundamental, because to accompany someone, you must be together with them, and we, if there is one thing we have learned, it is that only together can we emerge from this crisis in which we are daily forced to live, whether it is the...

****Crisis of the post-pandemic or returning pandemic, whether it is the crisis of rising raw material costs, the energy crisis, the war crisis, or the drought crisis. We can only emerge from this complicated situation if we work together, if we help each other, and we need the Institutions to help us, we need the State to help us... that State which was born here, which was born in this square where the first Senate of Italy was established. A value for us, not only romantic and passionate for the pride of our history, but also a value of perspective. Therefore, I believe there was no better place to celebrate your 25 years than to work here. We also have, on the subject of personal data protection, a particular attention because the Region, according to our Constitution, has competence over health and healthcare, and therefore we have the whole issue of managing people's health data. An aspect that I experienced first as a legislator a few years ago in the European Parliament, when I also approved the latest regulation on personal data protection, the one currently in force. And I found myself having to explain it, because it was evident that this brought a series of additional obligations and burdens for all those who, in their professional lives, had contact with personal data, data that had to be managed in a certain way. But I did it with conviction, because it was important to convey to public opinion that we were moving in the direction of protecting people's freedom.**

One thing that I believe would be appropriate to do today, and it would be appropriate for our government to work in this direction, is to explain more to citizens what the Authority does, to explain more to citizens that what may seem like a hindrance, what may seem like an additional bureaucratic obligation, is instead a protection of people's freedom. Sometimes regulation makes the President angry: in these two years of pandemic, I found myself, imagine, in the situation of having - according to the Italian Constitution - the responsibility for the health of my citizens, but not being able to know whether they were vaccinated or not. Imagine that paradox, but then, addressing the issue, we found with all the Regions of Italy, with the Conference of Regions, with which the Authority is in close contact, that balance of common work that allowed us, on the one hand, to protect what was the rightful protection of personal data, but on the other hand, to be able to intervene with those who are the competent subjects to also know personal data. But this is a theme, a reflection, that we need to bring more among ordinary people, we need to explain it more.

Earlier, Mayor Lo Russo recalled the research on the Internet. The regulation that we approved in the European Parliament was precisely the limitation of what they call cookies, that is, the fact that accessing a search engine records what you have searched for. In fact, the regulation imposed that one must say "I agree," but if you do not do so, the screen is blocked and it is not possible to continue the search. However, it must be explained to people that a minute later, if I search for a kitchen, that search engine knows that Alberto Cirio is looking for a kitchen, takes the data, and sells it to those who market kitchens. This passage introduces a marketing activity by subjects who, knowing my needs, then set their sales activities, but at the same time violates the freedom of the person. It is what supermarkets did first with loyalty cards. Few remember this aspect, but once the card is swiped, your purchases are recorded. And within that is our life. From my purchases, it is possible to know if I have elderly people at home, if I have children, if I have pets. It is possible to discover what my needs are. Today the system is changing. The Internet provides other tools and therefore it is right that there are other guarantees for citizens. But this, and I conclude dear Agostino, you said it well, quoting the European President, people must understand it. And we must be the first to make them understand

that this is truly a battle for freedom. Because there is no freedom if there is no protection of what are the personal inclinations of each individual, which belong to the individual sphere and which no one has the right to disrespect. In fact, they should not even know them.

There are aspects of the individual that must remain in the confidentiality of the individual themselves. Respect is the next step, because to respect you must already know, but we must take a step back: my personal identity is rightly my property and that of whom I want.

So, thank you for choosing Turin and thank you for the freedom that you transmit to us today.**

Greetings from the Deputy Minister of Economic Development

Sen. Gilberto Pichetto Fratin

Happy birthday to the Guarantor!

Thanks to President Stanzione and, of course, to the entire Board, to Agostino, whom I cannot help but imagine in the role of “member” of the Board and who for me continues to be simply “Agostino.”

I am pleased to be here to express appreciation on behalf of the Ministry of Economic Development and the national Government during this significant moment, also from a territorial perspective, because it is important to raise awareness at the local level about the role of the Guarantor and its relevant function.

Moreover, with the Ministry of Economic Development, as the delegate also for value papers, we have recognized the importance of these twenty-five years with the issuance of a stamp that symbolizes the State. The Ministry of Economic Development, in fact, has the role of recognizing the symbol of the State to those who deserve to be mentioned and to be remembered in history, beyond the postal function.

The task of the Guarantor Authority has already been outlined, so I will not say more. However, I must add that, reading the program, I noticed an evolution in the role: if perhaps thirty years ago we could talk about newspapers in relation to privacy, today I found terms like “metaverse” and “smart city”: these are the new frontiers.

The use of modern artificial intelligence tools risks being something incomprehensible and then unmanageable by the consumer. I say this also as President of the National Council of Consumers and Users, which must protect and stimulate the Guarantor Authority in safeguarding privacy at a particular moment for our Country and the world.

President Cirio spoke about it earlier: it is a moment that sees us in a historic change because we have had the pandemic and we do not know what the outcome of a new geopolitical framework will be. We must face inflation, which those under fifty know if they have studied it in school books, and therefore, there is a multiplication of initiatives alongside technological development that risks overwhelming particularly the most vulnerable.

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Thus, your function is also precisely to protect the most vulnerable, those who risk being left behind. And it is a challenge that we all must truly face, keeping in mind that it is part of our vision. So, once again, happy birthday to the Guarantor for Privacy, to the Board, and to all operators.

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Keynote Speech

- Dr. Agostino Ghiglia

Member of the Guarantor for the protection of personal data

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KEYNOTE SPEECH

Dr. Agostino Ghiglia

Member of the Guarantor for the protection of personal data

I thank the representatives of the Institutions: the Mayor of Turin, Stefano Lo Russo, the President of the Piedmont Region, Alberto Cirio, and the Deputy Minister of MISE, Gilberto Pichetto Fratin.

I thank the Parliamentarians and the academic and military authorities present, as well as all the "friends" of privacy who, for 25 years, have cooperated - together with us - in the protection of our data. I thank Dr. Pagella, director of the Royal Museums of Turin and the Royal Palace for the hospitality in the enchanting setting of the Royal Palace.

I thank the Guarantor, from the Executives to all the employees, for the enormous work done to protect our Freedom and on this occasion in particular the External Relations and Media Service led by Dr. Baldo Meo for the valuable contribution to the realization of this event.

I thank my staff (Cristiana Luciani, Licia Cristiano, Raffaella Bufo) for their tireless and intelligent support and, on this occasion, in particular Dr. Alessandra Genisio for her tireless coordination work.

Lastly, but only to emphasize with my heart the gratitude, I thank my exceptional Colleagues: President Pasquale Stanzone, Vice President Ginevra Cerrina Feroni, and my Colleague Professor Guido Scorza; our diversities find synthesis every day to do, paraphrasing Andrea Jelinek, President of the European Data Protection Committee: "the most beautiful job in the world: protecting people's data and with it their freedom."

And so "there will be a thinker who will build a brain that knows how to think exactly." It would seem that Goethe, in his famous Faust, had already...

1) Goethe, Johann Wolfgang von. "Faust," In Works, vol. 2, edited by Emilio Castellani, pp. 311-312. Turin, Einaudi, 1962.

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****Foreseen with a sort of prophetic anticipation, what is happening to our cybernetic society, increasingly abandoned, in a daily path now predominantly unaware, to technology. Lives that, mixing in an intrinsic bond of reality and virtuality, flow through the crowded streets of our cities and digital rivers of bits. However, it is not - and must not be - sufficient for man to merely be part of reality: man needs to know that he "inhabits" the world and how he inhabits it. The human being requires an environment endowed with meaning, a context in which everything maintains a certain relationship and whose evidence can indeed be learned, understood, explained. The human aspiration has never been to live in an environment closed off by biological determinism nor, even less, mechanical, but in a world constituted by a multifaceted reality: from the familial, working, cultural reality, to the affective, social, up to the national community to which we belong and, symbolically and materially, to Humanity as a whole and in its complexity. The hypersonic development of technology confronts us with exchanges that were unthinkable just a few years ago in the processes of interaction between man and machine. Man "machinizes" and the machine humanizes... The priority, the challenge of the coming years, therefore, is to humanize technology and not to machinize man, rendering him amorphous, a routine automaton that cedes too much of his life to machines. Technology, on the other hand, has always responded to man's natural vocation for action and discovery aimed at his vital and productive interests. We could then assert, from here to a few years, that artificiality is something better than naturalness and that its function - the function of machines - is to protect us from nature or to profoundly modify it, dehumanizing us? In the current and future digital society, the answer, which might seem rhetorical, is far from being trivial and taken for granted. If machines are to be tools at the service of man, it is essential that man, in using them, never loses control of his individuality and capacity for self-determination, that he never confuses the means with the end and that, above all, he does not delegate the end to the means.**

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Will we manage to remain aware of the necessity to consider the "technological" part of our being as subordinate, or will we confuse and gradually merge it with our humanity? Man, in making choices, recognizes a profound characteristic of his actions: good and evil. Man, with his free will, develops a sense of responsibility that we define as ethics. It is ethics, a human characteristic that makes us unique, based on values, but these values must in turn be based on knowledge which is, first of all, the

ability to understand and comprehend and not on a neo-fideistic acceptance of the future. With the advent of the digital age, this capacity for understanding must inevitably be supported by new forms of digital education; without it, the risk is that the "posse non peccare" that Saint Augustine placed among the foundations of free will will be lost. The machine also makes choices based on values, but these are numerical values, related to data. If we want, however, for the machine, without ever replacing the human being, to be of actual support to man and the common good, then algorithms must include ethical values and not just numerical ones, and this means that ethics needs to permeate computer science. It will be necessary to regulate principles and ethical norms in a language understandable and usable by machines, pursuing the study and codification of what Paolo Benanti, as early as 2018, defined as "algoritics." Only in this way will artificial intelligence be a revolution that leads to genuine development not only economically but also socially and culturally. In a word, a "humanly sustainable" development. Nevertheless, when discussing ethical values, ethical and moral limits - in this case related to the applications and implications of Artificial Intelligence - it must be prefaced that despite the existence today of widely shared and often recognized values in almost all areas of the planet, one cannot overlook the fact that cultural roots, and therefore moral values, differ.

****2)**** Benanti, Salvatore, "The Ethical Challenge of Digital Intelligence" in *The New Frontiers of Technology and Science. The Future Between Technology, Ethics, and Law*, edited by Mauro Nicolini, pp. 71-84. Aracne Editrice, 2018

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from country to country and observed and metabolized differently, especially where there are no legal norms that incorporate them or constitutions that identify them as values to be protected. We Westerners still have a nineteenth-century tendency to assume that our "Weltanschauung," our worldview, is "the" objective and objectifiable view, and therefore exportable. This is no longer true, even numerically. It does not seem therefore unreasonable to believe that an additional difficulty lies precisely in thinking of universal algorithms as the basis for future Artificial Intelligence; "value-based" algorithms accepted by a new "digital UN." In this regard, the European Union's choice to adopt a General Regulation on Artificial Intelligence (Artificial Intelligence Act of April 21, 2021) appears fundamental, but its final approval is increasingly urgent. The hope, or rather the urgent necessity, is that the ongoing discussion produces harmonization and legal certainty on which the reliability of technology use must be based, without obviously constituting a brake on progress and research. However, we are faced with another equally important need: decision-making speed; it may not seem paradoxical to say that the future urges us, pushes us to do well but to do it quickly.

Just as the GDPR initiated the global "revolution" of Data Protection (remember that it took more than four years of negotiation and six years before it came into force), the AI Regulation should hopefully become a global standard drawing influence and inspiration from the GDPR and serving as a catalyst for an international regulatory process. At stake is not only the pure certification of new technological tools for the protection of personal (and non-personal) data, for which the Regulatory Authorities must, in my opinion, position themselves as controllers and supervisors, but also the mapping of the coordinates on which to base a new model of society. A society in which human and non-human components will be called to coexist durably, according to an innovative principle of technological subsidiarity in which, as mentioned before, the higher-level entity remains the human being.

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The ethical-technological challenge we face, therefore, requires a cross-cutting commitment that, spurred by renewed and swift regulatory activism, involves not only the academic community but all those - Institutions and civil society - who care about human growth and development in a society that, while exploiting the enormous potential of new technologies, guarantees respect for the rights and fundamental freedoms of individuals. We must be aware that the ongoing digital revolution is not simply a technological evolution but a new discovery of fire that will, however, lead to an evolutionary timeline that is difficult to imagine. In this context, our Authority must assume a central role in defining

rules that, combining technological innovation and law, allow for the connection of the benefits offered by scientific progress with the guarantee of respect for the dignity and fundamental rights of the person. The complex of norms that guarantee the freedom of the individual (habeas corpus) needs to be renewed as "habeas mentem" in the face of new, daily, indefinable, and unimaginable threats and the possible conditioning that artificial intelligence systems can have on the mind of the individual in terms of decision-making freedom.

The algorithm must be circumscribed in its use. It is worth recalling Article 22, paragraph 1 of the GDPR, which states: "The data subject has the right not to be subject to a decision based solely on automated processing, including profiling, which produces legal effects concerning him or her or similarly significantly affects him or her" (unless exceptions apply for the conclusion of a contract or consent of the data subject). This provision, among other things, enshrines the right to human review of automated decisions and the prohibition of discrimination, already acts as a brake on potential invasions and predominance of the algorithm over the choices that must be made by humans. Consequently, data protection authorities already possess the necessary competence, human intelligence, history, and customary practices to ensure the best approach to the issue, minimizing the risks and criticalities that can undoubtedly arise.

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with the use of artificial intelligence systems. For this reason, Data Protection Authorities must be promptly and convincingly strengthened, equipping them with robust human resources in the technological sector to complement the more traditional legal areas.

So far, the use of information technologies has seen humans interact with the environment mediated by ICT systems - over which humans still had control - the ecosystem that is being established with the combined use of technologies such as the Internet of Things, Big Data, 5G, intelligent agents, and Artificial Intelligence systems imposes a radical paradigm shift: humans become users of the service provided by the system, partially or totally delegating to it the control of the interaction with the environment.

The use of technologies, in a digital ecosystem that sees Artificial Intelligence as the central element of the new technological paradigm, could become a formidable tool for progress in favor of humanity, even when it has the capacity to understand our sensations, our moods, our emotions, and our deepest impulses. If the answer is yes, in the face of emotional computing, it will be necessary to foresee protections to ensure the autonomy of the individual in the cognitive process before it becomes volitional.

Otherwise, the vast applications of Artificial Intelligence, even in its most trivial uses such as home automation or smart assistants, risk becoming an unconscious habit, routine gestures, passive and unaware profiling of attitudes, tastes, and emotions. When a large part of our choices can be captured by sensors, processed, and synthesized through machine learning, the framework of guarantees for the individual, evolving, must remain the solid foundation previously mentioned: a combination of ethical principles of processing and the fundamental principles of personal data protection - including legitimacy of processing, transparency, "explainability" of processing operations, data minimization, and security of processing - which constitute an indispensable reference for the protection of individual and collective freedom and consciousness.

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As historian Yuval Noah Harari says, in putting humans back at the center of the new narrative of the 21st century: "in a world flooded with irrelevant information, clarity is power." Clarity in remaining profoundly human and critical, not just lazy digital individuals, raised with a digital culture that, if it replaces the traditional one, risks "producing" a profiled and neo-massified society.

Undoubtedly, we are entering web 3.0 where virtual reality and augmented reality will allow us to "live" in a world where 3D virtual experiences will merge with physical ones. How can we not mention the Metaverse and the Ultraverses? In 2004, Facebook was born, and with it the "social age." Today, this tool (which still boasts 3 billion users) no longer reflects the tastes of millennials nor, even less, of generations X and Y, and so, over a year ago, Mark Zuckerberg launched Meta, not just a simple rebranding but a different idea: a virtual, augmented reality, in which our avatars through the Oculus headset will live, love, buy...

In this new "reality" - albeit distant due to the enormous management and interoperability problems between "worlds" it entails - it will no longer be necessary to use Photoshop: there will be digital beings, male or female, who will move and "live" in our place. To better frame the topic, let us recall two numbers: social networks have about 4.6 billion users, equal to 58.4% of the population; we spend about two and a half hours a day on social networks, which becomes over four hours in the age range of 11 to 18 years.

One might wonder if the Metaverse is already among (in) us. Meta, moreover, is just one of the 141 ultraverses so far recorded (at the time of publication of this book); among these are DECENTRALAND, where one creates their avatar and has the opportunity to buy and resell plots of land using Mana currency; EARTH 2, a metaverse where the entire planet has been divided into a grid of 10x10 meter plots to be sold to the highest bidders, on which one can build their commercial activity; FORTNITE, ... and so on with the imagination! The ultraverses will surpass the level.

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****Virtual, Immersive, and Augmented Games for the Few, Wealthy? "The Future is an Hypothesis" as Ruggeri sings.****

Despite the fact that the Metaverse is, to date, merely a trendy term that many discuss and few understand the meaning of or, better, the real potential and true state of implementation, it is worth noting the figures surrounding the metaverses (in 2021, there were indeed 500 million dollars in sales in the Metaverse, a figure estimated to reach 1 billion for the current year with a projected growth of 32% in the coming years until 2028).

However, such figures, which appear enormous to us, represent a small fraction compared to the billions of investments made by the platforms that create and manage them, with negative results that have thus far led to the dismissal of tens of thousands of workers engaged in the "metasector."

It is important to emphasize how the volume of data or metadata being generated, and that will be stored in the ultraverses, is infinitely greater than what can be acquired on the web, as we have known it until today. The user will no longer need to proactively provide personal data by accessing their smartphone and using one of the many apps. Rather, their data will be collected in the background while they carry out their virtual life.

The type and quality of the data obtained will also be very different: consider the possibility of acquiring the user's "gaze," the dilation of pores in predictive optics, or certain emotions and movements that the so-called virtual reality headsets will be able to collect. Not to mention the biometric data collected, which will inevitably require a profound rethinking of the verification of consent by users and, above all, minors.

At this point, is it reasonable to hypothesize a metaverse for adults only or, at least, for those over 16? This is a legitimate concern considering that the metaverses could create many new types of illicit activities and crimes, also—but not only—through the use of techniques for overlaying the human image based on Artificial Intelligence (so-called deepfakes), which will be even more manageable with avatars.

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A revision of the legislation on computer crimes dating back to 1993 will be essential. Closely linked to the above is the market for NFTs - unique, non-fungible tokens that are, therefore, irreplaceable and non-replicable and possess a digital value - with the possibility of conducting transactions in cryptocurrencies that will inevitably require ensuring an adequate process for managing anti-money laundering and anti-tax evasion procedures, thereby providing greater certainty to value exchanges that, moreover, given the instrument, are immutable and therefore definitive.

In conclusion, the metaverses and digital development in general pose revolutionary ethical and legal problems, further to those evoked above by the countless manifestations of AI. On the other hand, we have officially entered a new, radical era: the Digital Age. We already live in a double and parallel reality. Can we really think of going back or pretending to stop all this? If in this case the answer is no, a profound change will be necessary, starting from educational forms of digital education to ensure that new generations are masters and aware of their future, not mere consumers (of their personal and sensitive data) or avatars trapped in a virtual existence.

It does not seem excessive or science fiction to assert that the advent or predominance of the digital era—from SPID to CIE, from metaverses to “simple” e-commerce—represents an unprecedented epochal revolution (if not since the Lower Paleolithic, with fire...) that will change our way of living at an evolutionary speed that has no comparison in recent centuries.

It is important, in short, to create and spread awareness, sensitivity, and culture in the face of changes that the grandfathers Orwell or Asimov (yes, they are now very old too) would not even have dreamed of, so that Nietzsche's prophecy contained in the book “Human, All Too Human” does not come true: “All human life is deeply immersed in untruth.”

Will this be the challenge for the next 25 years?

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****First Session**

The Constitutions of the Metaverse:

Do Meta-Laws Occur?**

- BUILDING THE METAVERSE TOGETHER

Dr. Stefano Fratta

- HOW TO REGULATE THE METAVERSES

Prof. Francesco Pizzetti

- FROM THE METAVERSE TO THE METACITY: THE FUTURE OF THE NEXT GENERATIONS

Prof. Derrick de Kerckhove

- RELIGIOSITY AND SPIRITUALITY IN THE METAVERSE. A CONTRIBUTION TO KEEPING THE HUMAN AT THE CENTER

Don Luca Peyron

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****FIRST SESSION****

****THE CONSTITUTIONS OF THE METAVERSE:**

DO META-LAWS NEED TO BE ESTABLISHED?**

****BUILDING THE METAVERSE TOGETHER****

Dr. Stefano Fratta

Emea Privacy Policy Director at Meta

Since Mark Zuckerberg announced the new name of the company (now Meta) and his intention to contribute to the realization of the Metaverse, several topics once considered niche have become the subject of intense public and political debate. Today, we have a unique opportunity to discuss the opportunities and challenges of the digital revolution of Web 3.0. For the first time, the regulatory

framework will not have to intervene on a technology that is already fully realized, but can evolve concurrently with its development.

If we were to give a general definition, we could say that the Metaverse is a collection of interconnected digital spaces accessible through a multitude of different devices and technologies: augmented reality glasses, virtual reality headsets, personal computers, smart displays, and mobile phones. The Metaverse has the potential to make our online experiences more realistic, allowing us to go beyond the two dimensions of today's Internet.

The Metaverse and immersive realities are enthusiastically welcomed in some circles and approached with skepticism in others. However, it is easy to understand the reason for this, especially when society is already questioning how technology operates in today's two-dimensional world.

When Facebook was born 18 years ago, online navigation was mostly limited to consuming and drafting written texts. With the advent of smartphones, equipped with increasingly sophisticated cameras, photos have taken on a central role in digital interactions, communication has become increasingly visual, and the Internet has increasingly relied on images.

Moreover, as connections have become faster, videos have established themselves as the preferred means for sharing and communicating daily life. The Metaverse, therefore, is nothing more than the next natural step in the technological journey made so far: an experience that will no longer be two-dimensional, but immersive and, consequently, much more engaging and realistic.

Three key factors will make interactions in the Metaverse more similar to those of our daily lives: namely, ephemerality, physical presence, and immersiveness.

Ephemerality will be a fundamental characteristic of our interactions in the Metaverse. In the physical world, almost all the conversations we have are "ephemeral"; that is to say, there are no permanent records of what we say in our daily lives. In contrast, internet searches, emails, SMS, and written posts on social media are characterized by a more lasting permanence. In the Metaverse, there will be a shift towards oral and direct communication, just like in face-to-face conversations.

And if nothing so far has reached a level of audiovisual perception capable of recreating the feeling of sharing spaces, as happens during a conversation with friends at a bar or a meeting with colleagues in the office, the immersiveness of the Metaverse will allow us to communicate with others as if we were actually together with them.

The goal of immersive technologies, however, will not be to replace in-person gatherings. On the contrary, the Metaverse could establish itself as a complementary space that will make it possible to create communities and maintain relationships even when various limitations prevent in-person meetings.

We can therefore imagine the Metaverse as a building where each floor supports the one above it. The foundations of the building include hardware (phones, virtual reality headsets, augmented reality glasses, etc.), protocols, and technical standards that ensure interaction between various technologies—in jargon, their "interoperability." The ground floor of the Metaverse will consist of networks and platforms that realize the 3D worlds of the Metaverse in their entirety. The first floor, on the other hand, is populated with experiences that each user can access.

For example, users of Quest, our VR headset, can access virtual experiences that enhance the online work experience, such as Horizon Workrooms.

In this context, augmented reality allows for the combination of reality and the virtual world, enabling the visualization of virtual images overlaid on the view of the surrounding environment through the use of glasses equipped with transparent lenses. Through virtual reality, on the other hand, we will be able to immerse ourselves in a completely digital three-dimensional environment, thanks to the use of specific headsets. Overall, the technologies underlying the Metaverse will allow us to communicate, stay connected, and keep in touch in a simple and engaging way, thanks to the development of holograms and avatars.

The common thread in this discussion is interoperability, that is, the interconnection of standards, systems, and apps that allow people to travel easily from one part of the Metaverse to another. This is not an absolute: not all elements of Metaverse experiences need to be, or will be, compatible with each other. However, without the integration of a significant degree of interoperability at every level, the Metaverse will result in a fragmented and divided space with impenetrable compartments from one

another.

To accelerate this process, the Metaverse Standards Forum has been established. The Forum will explore the points where the lack of interoperability hinders the spread of the metaverse in a collaborative space involving founding members, including Meta, as well as Adobe, Alibaba, Autodesk, Epic Games, Google, Huawei, IKEA, Microsoft, NVIDIA, Qualcomm Technologies, and Sony Interactive Entertainment. The development of the metaverse is therefore a collective process, a path towards the development of a governance system that involves the private sector, legislators, civil society, the academic world, and the people who will use these technologies.

AR and VR have the ability to change the way people interact with each other, and for this reason, it is crucial that the development of these technologies is responsible and transparent. The pace of technological change is very fast and can raise questions or concerns, and it is therefore essential to discuss risks and opportunities with all stakeholders involved in the development and regulation of the metaverse, such as institutions and regulatory authorities. Our commitment is aimed at improving the understanding of these revolutionary technologies because we believe it is the best way to foster innovation and enable everyone to benefit from it.

On one hand, Meta has launched, together with Essilor Luxottica, an awareness campaign across various media to improve social acceptability and awareness around the use of smart glasses, focusing on explaining how to use these tools responsibly by the general public. On the other hand, we are committed to ensuring that reflection on the regulation of immersive realities begins today. There is nothing deterministic about how a technology interacts with society. Technology, in itself, is neither good nor bad. Therefore, its impact depends on how it is used, and this is why it is necessary to discuss together how to ensure an appropriate development of the ecosystem, drawing lessons from the development of the existing one concerning the Internet and social media. However, the rules cannot be identical, as in many respects the experiences within the Metaverse will be more similar to those of physical reality than to the Internet as we know it today. Furthermore, the rules must be flexible and able to adapt to a technology that is still evolving.

Let us think about privacy protection. How can we minimize the amount of data used, build technologies to enable their responsible use that guarantees privacy protection, and offer people more transparency and control over their data? Or, we can reflect on data related to human characteristics. The processing of data generated by the human body is fundamental to the nature of the metaverse. One of the main issues is to establish a clear definition of this data, which we might call Human Characteristic Data. Although it is not a new issue in itself, it is crucial work for the future of the metaverse.

Human characteristic data includes "physical, physiological, and behavioral data related to tracking hands, body, face, head, or eyes, typically positions, gestures/movements, gaze." The processing of this human characteristic data represents the essential premise for the functional experience of social connections in the metaverse, which is not aimed at uniquely identifying an individual but rather at enabling human communication and social interactions through other means.

Let us also consider the topic of digital identity. Users will have the opportunity to create different distinct and separate identities on platforms and thus experience new identity experiences through physical avatars. We believe this offers people a great opportunity to express their authenticity through various iterations or representations of their digital identity in the metaverse. In this regard, there must be collaboration between companies, experts, and policymakers to define shared standards regarding fundamental issues such as identity, privacy, and security.

We are just at the beginning of our journey towards the metaverse. Many of the experiences will only be accessible in 5-10 years, as will many of the use cases we envision today.

This timeframe gives us the opportunity to ask the most complex questions about how these products should be built and offers our company, and all companies involved in innovation, the time to invest in research, work closely with regulatory authorities and policymakers, and to better understand the challenges and opportunities of these technologies even in the early stages of their development.

HOW TO REGULATE THE METAVERSES

Prof. Francesco Pizzetti

Full Professor of Constitutional Law at
the University of Turin, former President of the
Italian Data Protection Authority

First of all, I would like to express my heartfelt thanks for the invitation to this meeting on such an interesting topic. There is no doubt that the perspective of the Metaverse is set to open a new phase of digitalization and thus change the way humans live. Moreover, as is always the case with changes of this magnitude, it is extremely difficult to understand what the effects will be until the new technologies are fully operational and widespread. During the pandemic, we had a concrete demonstration of how technological changes can have, depending on the context in which they operate, even unforeseen effects. One only needs to recall the enormous impact that distance learning, made possible by digital technologies, had on the lives of young people during the pandemic to understand that the effects of innovations can be of such magnitude as to bring about permanent changes in habits and in the lives of human groups and in the relationships among their members.

It is certain that today, several months after the peak of the pandemic and as medical risks change and diminish, it remains extremely difficult to organize in-person conferences, as the use of technology to organize remote meetings has now spread in a way that seems irreversible. However, thanks to the web, these meetings are also substantially in-person, allowing participants not only to see each other but also to interact and discuss in ways very similar to traditional in-person conferences. The effects on the organization of educational activities have been even more pronounced, leading to the not-so-pleasant acronym DAD to refer to educational systems that develop entirely online, including exams. This has, among other things, created a challenge that is still ongoing between traditional universities and online universities, resulting from the fact that even traditional universities are increasingly organizing educational activities that rely on telematic technologies. It is therefore quite difficult to say now what the effects of the Metaverse will be, not so much in relation to the use of the internet but regarding the types of human relationships that, using this new technology and thanks to it, may develop.

It is therefore quite clear that in such a context and in the presence of such an important change, announced some time ago and to which impressive resources are being dedicated worldwide, there is a strong temptation to begin exploring the future in order to try to contain its most risky effects for our societies and their way of life. This, after all, was the broader significance of the aforementioned intervention by the representative of Facebook. It is an intervention that I consider very important and which, for this reason, deserves to be examined with due attention. The core of what we have been told seems to me to be the assertion that we are certainly living in an era of great changes, regarding which a sense of realism and correctness must lead us to say that we do not know where we are going and we do not know what evolution society will undergo.

It is an ongoing process that recalls the journey of Christopher Columbus, who believed he would find the route to the Indies and instead found the Americas.

We too do not know what we will find and where we will land within this digital evolution of society, which certainly began several years ago. Moreover, it is worth keeping in mind that the Internet was initially conceived for military purposes and to be at the same time an easy tool for global communication, characterized by specific features such as being easily expandable in space, easily modifiable over time, easily repairable, and responsive to the need to ensure simple and secure military communications within NATO.

Let us also never forget that the Internet is the very foundation of the digital society; it was not a product of the economy, although its evolution has certainly favored the development of technology and global economic relations based on what was once defined as the Washington Consensus: a doctrine that developed thanks to the thinking of J. Williamson in the late 1980s and revolves around 10 directives.

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formulated by Williamson, is primarily based on the belief that technology and the economy, when left free to operate without excessive constraints and legal conditioning, are more capable of producing innovation. This is a very important vision that has also dominated the thinking and activities of the EIB, the International Monetary Fund, and major global financial institutions. We talk about it a lot, perhaps too much, often forgetting that it has greatly accentuated the difference between Europe and the United States, going well beyond the diversity that we usually see, especially related to the protection of personal data. The reality, however, is that in the digital society, the difference between the USA and the EU is much broader.

Especially from the phase following the entry into force of the GDPR, Europe (EU) conceives the regulation of technologies also as a tool to recover the delays accumulated by the European Union in global technological competition. This gives rise to the vision of greater regulation as the ability to set rules that allow for the construction of a more protective, orderly, and safeguarded digital society, in which citizens can operate and feel more protected, even well beyond privacy as we traditionally understand it in the EU. The European vision is precisely the opposite of the American one, and the clash is much deeper than what is told or perceived.

Unfortunately, even in scientific and technical communication, we all live in an era that is not excessive to define as “spectacle communication.” Communication is increasingly less about informing and comparing the results of research and scientific analyses. Instead, information broadcasts are increasingly dominated by the desire to gain visibility, notoriety, and above all, at least for broadcasters, audience. This evolution is even stronger in ordinary communication: information as a tool for mediation between public opinion and the phenomena that occur is increasingly diminishing. On the contrary, spectacle communication is increasingly asserting itself, also for obvious economic reasons. In fact, it costs much less for an hour of television broadcast done by a journalist turned anchorman who invites 3, 4, or 5 experts who then gain notoriety with all that follows, than it does for an hour of entertainment based even just on a group of dancers and an orchestra.

In this era, it is becoming increasingly difficult to explain what happens to public opinion, and it is becoming increasingly difficult for the public to understand and comprehend what they hear or read in the media. However, it is certain, and easily verifiable, that the European Union has set itself on a path since 2020 that sees it as a clear antagonist not only to the Chinese model, which is essentially based on control technology, but also to the American model, which is based on competition among service providers and antitrust controls. The EU, in fact, prefers to regulate network technology and provide guarantees to citizens through the regulatory framework of the digital society rather than limiting itself to antitrust controls to regulate a digital market seen as a challenge rather than as a social Eldorado.

This aspect of the EU strategy is the basis of the entire package of regulatory proposals known as the Digital Services Act package proposed by the Commission since 2020 (DSA) and 2022 (DMA), on which an agreement was reached with the Council between March 22 and April 23, 2022, so much so that it is expected that the proposed measures may come into force by 2024. These are two very demanding regulations that confirm the continuous production by the EU of regulatory proposals, including that on artificial intelligence presented on April 21, 2021. Proposals that all aim to strengthen, first and foremost, the single market, ensuring uniform regulation throughout the EU territory and therefore, through the so-called “Brussels Effect,” also conditioning the global digital market or at least a large part of it.

The digital evolution in the United States is different. In the USA, the digitalization of the economy and society has continuously intensified since the Clinton directive, which, also influenced by Al Gore, opened up the commercial use of the network at a time when even the new web protocol was profoundly changing the potential of digital. All of this placed the USA in a perspective oriented towards leaving the

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as free a market as possible, based on the belief that it would be capable of ensuring, through its own functioning, the maximum possible innovation. At the core of Clinton's approach, which later became stable in the U.S. vision, are consumer users who drive the market and push it to develop ever-new technologies capable of meeting their needs. It is undeniable that the American vision of the Clinton era applied to the internet has profoundly marked its characteristics and has accentuated its continuous, constant, and accelerated development, as evidenced by the rapid evolution of the digital society in the United States since that time. Indeed, that was the era of the development of Silicon Valley and social media, which, also thanks to the development of Web 2.0, began to colonize the internet and thus the world with the services offered and the commercial use of collected data. Essentially, since the transition from the predominantly military use of the internet at the turn of the late 1990s and early 2000s to commercial use and the development of Web 2.0, a prevailing orientation in the U.S. has been based on the belief that simply having trust in the market, overseen by antitrust control, would ensure competition in digital technologies and thus also their rapid development in protecting consumer interests.

This has, of course, further accentuated the gap between the U.S. and the EU and has been—and still is—one of the reasons why the EU has always viewed the transfer of data to the U.S. with suspicion. Moreover, there are recent signals that even in the United States, the vision of the internet and digital regulation is partially changing. In the U.S., in fact, the regulation of commercial relationships, left predominantly to the States, risks, in the absence of a unified federal regulation, fragmenting the internal market no less than what was happening (and can still happen) in the EU. Hence the reason why the United States is now also reflecting on the risk of seeing its internal digital market fragmented by the different regulations adopted by the States, including those aimed at protecting personal data, or by the different needs for the use of digital technologies in various economic areas. It is no coincidence that recently the discussion of a federal data protection bill has begun (this is the American Data Privacy and Protection Act, commonly referred to as ADPPA). A proposal presented based on a justification that also concerns the digital society as a whole in a vision very close to that of the European Union. In fact, even in the U.S., there is a recognition of the need to avoid excessive fragmentation of the federal internal market: a risk that, as the EU case demonstrates, increases with the growing use of digital technologies. Essentially, it seems that in the U.S., the idea that the market is, by its very nature, a producer of innovation that respects the rights and interests of consumers is destined to increasingly go hand in hand with the different and opposing idea that unregulated innovation can fracture the internal market. All of this allows us to think that in the coming years, the difference in approach between the EU and the U.S. may lessen, accentuating, thanks to a rapprochement between the two approaches, future changes in the overall framework. This, of course, would also impact the regulation of data transfers between the U.S. and the EU. Within this framework, already quite complex and constantly changing, we must also keep in mind that the digital society has begun to gallop and technology is in continuous development.

We cannot predict today what evolutions this society will have: partly because, as I said, the cultural framework regarding the digital that dominates in Western society is driven by the American vision and not yet by the European one; partly because no one knows what questions will be posed to technology or what suggestions technology will make available to us. We can only be sure and aware that the future will be very different from the current situation, to the point that, as Floridi says, the children born in the coming years will ask us in astonishment how life was (and how it could be lived) in a fully offline society. However, precisely because of our underlying culture, which, especially in the last three centuries, has heavily invested, with excellent results, in science and technology, we are tendentially open to thinking that what innovations science is researching and proposing will certainly be useful for us. Only recently, especially in connection with the

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We have begun to seriously concern ourselves with digital evolution, but for at least 2 or 3 centuries, the secular world has bet on science, even at the cost of hard clashes with religious thought,

particularly with Catholic thought. The Enlightenment was this, and that is why Western culture has often clashed with those who wanted to curb scientific evolution in the name of the centrality of man.

Not by chance, today, to mitigate our concern about a technology that evolves too rapidly, we have placed back at the center the concern, more than respectable, for human dignity and therefore we have begun to increasingly value an ethical approach also regarding digital technology. We hope that this rediscovery of ethics is primarily a rediscovery of the necessity for human beings to reflect on the development of technology and not just to accept, blindly, as good and positive any technological innovation that science proposes to us.

That said, however, we must recognize that currently we do not even know what points of orientation to use precisely because for three centuries we have not posed this question to ourselves. Our attention has been focused only on the need for the safety of the technical means used. Consider, for example, how attention has been directed towards seat belts, airbags, or guardrails. Every innovation has been examined essentially not to avoid but to protect against the negative consequences that a bad use of it could have in a framework seen as positive nonetheless. If, on the other hand, the call to ethics today, in this phase of our society, is a push to return to reflect on the risks, as well as the advantages, of scientific discoveries, then it is certainly a good and positive thing but one that requires a very significant effort, including cultural effort.

However, I would not want to diminish the importance of the intervention of the representative of Facebook-Meta who tells us: but why don't we do the opposite of what has happened so far? In the years behind us, technology has advanced on its own, and then the regulatory Authorities intervened to tell us what was wrong, what needed to be corrected, what could not be done. If I understood correctly, today's intervention proposes to bring the Metaverse to life "together."

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Facebook and Meta would provide the technological expertise, honestly indicating the objectives that companies want to achieve: objectives that at the current state are not very clear also because the tools that should allow us to live in the digital world in ways very close to those of the offline world are not very clear either. However, precisely for this reason, the offer is even more attractive. Essentially, we are told: considering the complexity of the technology to be used, let's build it together—us technologists, you "ethicists," and you Authorities and regulators who safeguard the rules related to the protection of the fundamental freedoms of human beings.

As I have already said, this is a very interesting suggestion but, at least to me, it seems incompatible with the GDPR, and I believe that Meta knows this very well. The GDPR is based on another bet and on another pillar: what we call, perhaps not always in a sufficiently conscious way, "accountability." The GDPR essentially states that, especially within the framework of privacy by design and privacy by default, the problem of protecting fundamental rights falls entirely on those who design, build, verify, and use a new technology. However, this is supplemented by the theme of risk assessment, which falls entirely on the data controller.

In essence, it is up to those who are designing the technological innovation to evaluate all of this and in this framework the technology itself and the risks it may entail, including the responsibilities of the manufacturers of devices and those who make them available to the data controllers. However, it is up to those who use these technologies to assess the risk that their use may pose for the data subjects and their data and to adopt the necessary measures to reduce or eliminate such risks as well as to ensure that data subjects receive adequate information regarding such risks.

Based on this reflection, it is important to invite those who engage with the world of Meta to always remember Article 5 of the GDPR and what it says about the concept of data and the responsibility of those who, processing data, are required to comply with this regulation.

Therefore, certainly if in the future the Authorities of individual States or the EDPB wanted to accept the offer to study together the problems that Meta poses, it is from Article 5 that we should start.

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Anyone wishing to venture down this path must know from the very beginning that it is necessary to study and implement technologies that respect the constraints, conditions, and characteristics that must characterize these treatments under the GDPR. Constraints that have not coincidentally been summarized in this Article 5 of the GDPR, which is lengthy but not excessively so, and therefore substantially manageable. It does not depend on me but on President Stanzione and all his colleagues both in the Italian College and in the EDPB, composed of representatives from all European authorities, to evaluate and decide whether to delve deeper and agree to dialogue with Meta based on this suggestion and this regulation.

I will make just two observations and then conclude: be aware that the proposed path is indeed suggestive but presents aspects and risks that must make everyone reflect. We must absolutely avoid taking the regulation of technology out of the democratic sphere, especially when it concerns the protection of individuals. At this point, it could indeed be said that if this proposal is accepted, it creates a situation in which scholars of technological innovation, businesses, and the regulatory Authorities become in some way co-decision-makers who, precisely for this reason, tend to replace the political decision-maker and become very self-referential.

Therefore, this is a decision and a choice that cannot be made lightly. In my opinion, accepting it, however appealing it may seem, risks undermining a large part of the guarantees offered by the GDPR to the interested parties. If, in fact, upon testing Meta's functioning, the decisions adopted in agreement between the Authority and Meta were deemed by one or more interested parties to be in conflict with the GDPR, to whom could those individuals turn to protect their rights? And, in any case, what degree of actual and legal freedom could the Authority claim to deliberate, possibly even in contrast to the choices made by the majority of Authorities in the EDPB?

And again: how could the Authority participate in the work of the EDPB on the subject while objectively being in a very different position from that of the regulatory Authorities of other States that had not accepted Meta's offer?

Finally: precisely because of the characteristics of Meta, a completely new technology of which we can only intuit benefits and risks, how is it possible that the Authorities, by participating in the phase of elaboration and implementation of the project, risk compromising their essential role as controllers and verifiers of the actual functioning of the innovation we are discussing?

The issue raised by Meta's offer, which was reiterated to us today by its representative, compels us to always remember that we are a generation (and even more so will be the one that comes after us) called to face a great challenge. For this reason, in the face of the problems that Meta will certainly pose to the Authorities, it is necessary to renew our support, affection, and friendship to the Regulatory Authority and to those who currently bear the responsibility.

The Authority is, and must remain, the only true and effective protection of our fundamental rights in the face of the explosion of new technologies. In fact, both at the national and European levels, it is the only Authority that has accumulated a truly wide-ranging experience regarding the digital evolution of society for 25 years, both from an ethical, technological, and legal standpoint.

Alongside the Authority, however, there are national and union decision-makers who represent the European peoples. They cannot and must not renounce their duties and exercise their powers. The issues that arise with the transition to the Metaverse are enormous and still only barely glimpsed, but it seems clear to me that it is up to European decision-makers and European Authorities to say whether it is permissible for a person in the Metaverse to present themselves as a woman while being a man or vice versa; as an adult while being a minor or vice versa; or if, participating in a meeting of children, they can adopt an avatar resembling a child, which can obviously also become a tool for misleading interlocutors.

These are just a few problematic hints, yet useful to enable us to understand how strong the evolution

of digital society and, more generally, of our living reality in the Meta world can be. We will face new enormous problems that it is not right to think can be resolved by favoring or soliciting a prior agreement between Authorities and businesses.

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****On the contrary:**** it is precisely in the face of these evolutions and the problems they entail and will entail that we must increasingly strengthen the Authorities, safeguarding and enhancing the cultural and ethical role they play. A role that certainly has a political significance of the utmost importance, as demonstrated by the fact that both the DSA and the DMA imply a strong reinforcement of the role of the Commission, both in the evolution of the legislation and in the control of its compliance. In short, we have truly entered a "new world," and Meta will likely be destined to be its most evident signal.

However, one thing deserves to be strongly emphasized: the ongoing changes and those on the horizon increasingly demonstrate the necessity for new forms of guarantees for the freedoms and fundamental rights of European citizens and of people around the world. In this sense, we can be proud that the EU has anticipated with the GDPR a regulation that is proving to be more and more suitable and adequate for the digital society and, above all, that the EU, also counting on the Brussels effect, is trying to make a fundamental contribution to the construction of an increasingly digital planet that is also increasingly respectful of human essence and human rights.

It is therefore appropriate to conclude this intervention not only with a thank you but also with a strong wish to the Data Protection Authority and to those who lead it, expressing our full confidence in their ability to collaborate in guiding the Italian and European people with a steady hand into the new era that is opening before us.

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****FROM THE METAVERSE TO THE METACITY: THE FUTURE OF THE NEXT GENERATIONS****

Prof. Derrick de Kerckhove

Scientific Director of the TuttiMedia Observatory and MediaDuemila,

Professor at POLIMI, former Director of the McLuhan Program in Toronto

The 25 years of privacy that have accompanied the rapid evolution of our humanity have brought us to this point today to discuss the new stage of digital transformation, which is much deeper than most people seem to be aware of. I want to remind you that all change starts from the concept related to duplication. Before the digital era, duplicating something was complicated and expensive. Duplication, after duplication, has led us to the creation of the metaverse, which practically establishes our existence in three spaces. We are well acquainted with the physical space and the mental space, and then there is the virtual space. The point is that these three spaces are becoming increasingly less distinct from one another. Wearing a headset and entering the metaverse is simple, a game that takes us into a parallel reality where being and non-being blur, and the wonder of the digital world begins, with all its positive and negative aspects that the speakers before me have well illustrated.

But the metaverse is undoubtedly the new stage of digital transformation in its complex redefinition of space, because it is leading us toward the externalization of our imagination.

It becomes clear that the solid points of being human, namely space, time, and self, are inexorably and profoundly changing. In the 1990s, I anticipated the externalization of our memory; I spoke and spoke again about our technological prostheses: computers and then phones. Today, my vision of change is completed with the imaginary becoming real: there is no longer a screen that divides. The transformation brings us inside. Virtual reality is the opposite of a book, as it does not nourish internal imagination but presents our internal thought to the external world. In my view, the augmented reality of the metaverse reduced to simulation endangers our very human culture and also presents us with a political dilemma because a public good, by virtue of augmented reality, ends up becoming private.

I wonder what will happen when, as Matthew Ball (CEO of Epyllion, Venture Partner at Makers Fund, Senior Advisor at KKR, Senior Advisor at McKinsey & Company, author of the bestselling book **The Metaverse and How it Will Revolutionize Everything**, published in July 2022) says: "Today's generation of children often expresses, learns, and socializes constantly through virtual worlds that they can touch, modify, and collaborate in. This will not stop. In fact, the capabilities of these virtual worlds will expand, their ease of use will improve, and their importance will grow. Moreover, the 'iPad native' generation (or perhaps 'virtual world native') will continue to mature. Most of them are still consumers, few are creators, and almost none are business leaders. They will be. And their frames of reference will lead to transformative change."

Well, in this transformative change, how will we bring back social and legal rules, and above all, the concepts related to our ways of living and operating, such as that of privacy? The rules of living are tied to the sharing of language, which is currently under attack because the epistemological crisis we are experiencing has put it back in...

****Discussion of the Foundations.**** In fact, our alphabetic culture gave man the certainty of the relationship between language and reality. Reality changes, transforms, and reforms in the metaverse. Western man is part of the anthropological group defined as naturalism, which is based on participation and sharing of a common space (nature, indeed), and on the internal privatization of thought. How to manage privacy in this new context, the metaverse, is the challenge that I see ahead, especially for the Authority established 25 years ago to guarantee our private existence. I absolutely do not want to demonize; we cannot go back, but we must commit to understanding how it works in order to be able to respond to the challenge that this new space presents. Professor Pizzetti has already addressed the commercialization of this new space, which is private, but in a certain sense raises some issues, among these:

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public interest because it is available to everyone. And here I emphasize that there is not only the metaverse of Meta. After analyzing the point to which digital transformation has brought us, my solution is to support meta-cities where we can import some privacy rules, for example. In this change lies our future because the smart city in its metaverse version is based on the collection of data related to every aspect, from traffic to pedestrians, from crime to education, and the list goes on. From this practice depends the fear of losing privacy, which is a cultural issue. The United States tolerates flexibility in favor of "surveillance capitalism," while the European Union defends itself with the GDPR, and China chooses quickly and drastically: no privacy, period. What do we do? The answer to the Authority that has governed this aspect for a quarter of a century and which has been represented by a great friend of mine, Stefano Rodotà, with whom I have discussed several times about "is privacy over?"

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****RELIGIOSITY AND SPIRITUALITY IN THE METAVERSE****

****A CONTRIBUTION TO KEEPING THE HUMAN AT THE CENTER****

Don Luca Peyron

Digital Apostolate Archdiocese of Turin, Catholic University

The expression "human at the center" has always been rather ambiguous and risks being wishful thinking to the extent that it does not become more specific or does not offer greater criteria for specification. Theology and the vital experience of religions can offer an additional perspective, not necessarily alternative, indeed rather complementary, which, in dialogue with knowledge, provides a substantial contribution to the endeavor. The endeavor to safeguard the human in the digital metamorphosis and specifically in the presumably further metamorphosis that the metaverse or metaverses will offer or determine(1). The Catholic Church, despite its various and well-known flaws,

can certainly rightfully adorn itself with that expression with which Pope John XXIII iconically defined it: "Expert in humanity." And the Bible, the principle and foundation of religious experience and theology, is indeed a grand narrative of the human, in dialogue with itself and the world, in dialogue with God and the inner world.

From here arises the interest, or better said, the service that theology and faith can render to humanity, to believers, to believers in a different religious system, and to non-believers. The human that unites us, even if it does not always bring us together. Indeed, upon closer inspection, the issues arising from digital transformation and those connected to the metaverse are fundamentally questions of:

1) I will not dwell on the definition of metaverse or metaverses as this is the subject of other contributions contained in this collection of proceedings. It is worth remembering that the term "Metaverse" was coined by Neal Stephenson in the cyberpunk culture book "Snow Crash" in 1992. It is described by the author as a kind of shared virtual reality via the internet, where one is represented in three dimensions through their avatar.

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anthropological character. From them arise eminently technoscientific questions but also physiological, psychological, economic, legal, and social ones. But the human being, the idea we have of it, the sense that determines being, is the great key, the measure of reading and judgment, and, from our perspective, the ultimate goal of every human action, including technical action, I would dare to say especially technical action in this time. Let us pause for a moment on an overview from which to generate the reasoning I would like to propose by analyzing two structural limits of the human being, of each of us.

Upon closer inspection, the technological artifact has always been born to respond to a primary limit that insists in the very nature of humanity. Unlike other living beings on this planet, the human being is not able to survive in the nature in which it is born as it is, *sic et simpliciter*. A newborn baby does not need, like other cubs, only the care of its parents; it needs to be covered with the skins of other animals, to be defended from predators with sticks and spears, to be fed for several years before it is able, independently, to defend itself and procure food. The human being, in nature, succumbs with its only natural endowments. It does not have particular hearing or smell. It is not fast, does not fly, nor does it climb with particular skills. But it is intelligent, endowed with a combinatorial and creative intellectual capacity, which enables it to solve problems, create objects, learn skills, adopt completely new behaviors that are not inherited from parents, etc. With such intelligence, it compensates for the apparent gap that separates it from other creatures living in nature. And it surpasses them. It does so, to return to the starting point, with technique. It builds the stilt house, sets aside effective tools, forges others, gradually until one day it lands on the moon or puts into orbit satellite detection systems capable of mapping its world. It does not yet predict earthquakes and faces a pandemic in times that stretch out. But technique makes the difference, deceiving nature very well.

It is no coincidence that the word machine derives from the ancient Greek *mecainomai*, which literally means to deceive. And to deceive with almost divine appearances. It is worth recalling here the *Deus ex machina* of Greek theater, which later became a common expression. A *mechanè* (*mēkhanē*) was a sort of crane used in Greek theater, particularly in the 5th and 4th centuries BC. Composed of wooden arms and a system of pulleys, this theatrical contraption was used to lift actors into the air, simulating flight. It was certainly capable of lifting at least two people and transporting them to the middle of the orchestra or above the *skené*. For this reason, the *mechanè* was often used to simulate the intervention of a god on stage, to resolve issues so intricate that they could not otherwise be solved. From this tradition arises the Latin expression *Deus ex machina*.

Technique, therefore, is a typical expression of the human, an integral part of its nature. We are, to put it in other terms, naturally technological. This, from a theological perspective, is a good thing; it can be considered an accessory and quite indispensable good, granted by the Creator to the creature and a

full expression of its being in the image and likeness of God. Two examples can support these assertions. The first is the use of technique by God to save the righteous from the flood: the Ark of the Covenant is a technological construct whose plans are given by God Himself to Noah so that he could build that artifact (Genesis 6:13-16). Technique is thus part of the salvation and care that God has for His beloved creature. This horizon is reaffirmed in the earthly story of Christ, the archetype of the fulfillment that God has in store for humanity. Christ was a maker of technique, not just a simple carpenter, but something more. The Greek term we read in the Gospels of Mark and Matthew can indeed be interpreted in various ways. The word used in the evangelical texts is *téktōn*, used for artisans and woodworkers (and thus can be translated as "carpenter"), but it is interesting that it can also refer to stonemasons, builders, and even those who excelled in their craft and were able to teach it to others. The Latin translation we find in the Vulgate, *faber*, maintains the various meanings of the Greek *téktōn*. The carpenter, not just the carpenter of Nazareth, is a man, true God, who for a good part of His existence takes care of His fellow beings by constructing artifacts that allow them to live better. The table, the chair, the ladder that Jesus builds are not simply aimed at living, at having an income with which to live.

Every gesture made by Christ has a broader theological significance beyond the simple act itself. Thus, the repeated gesture of creating technique for humanity has a theological scope that is indeed far greater, and not yet fully explored by theological studies themselves. We can therefore conclude that technique, an expression of the human, is at the service of its being in nature and in its living in respect of itself and the nature in which it lives in balance.

However, the human being has a second limit that is not simply of a natural biological, physical, creaturely order. We could define it as interior, deep, spiritual, and psychological. It aspires to a fulfillment, to something more than mere living, feeling a chronic insufficiency in its deep being. That radical aspiration leads it to create art, poetry, music, and philosophy. That something more leads it to look at the sky not simply in terms of its material needs, but as inspiration for its spiritual, transcendent needs. The human being is in search of something more and different from itself.

A first response to this deep desire is found in the other, in other human beings. The Bible narrates this desire with the encounter between the masculine and feminine, in that diversity created precisely because the individual is not sufficient unto itself and because from that encounter arises the intuition that the definitive, totalizing fulfillment is in that totally other which is God Himself. This further passage, this necessary leap, is significantly evoked in another passage of Scripture where, in giving Moses the ten words,

**What we commonly call commandments,
binds to one of these,
and to this one only, a promise. It is the fourth
commandment, the fourth word,
which in the book of Deuteronomy is referred to as follows:
"Honor your father and your mother,
as the Lord your God has commanded you, that your
days may be long and that you may be
happy in the land which the Lord your God is giving you."
(Deuteronomy 5:16) Of the ten
words, therefore, this is the only one to which a
promise is associated. All the other
words, the other commandments, seemingly
even more important or
significant than this one do not promise anything, here yes.
For what reason?

The scope of this word is not what we
get from a first
reading or a reading marked by habit or by

cultural practice. The question at stake is not simply about treating parents well, respecting them when we are young and taking care of them when they are old. There is much more here. This is revealed by the Hebrew verb underlying what we translate as 'honor'.

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The term used is kabod, kabed (or kaved) which means give the right place, the right weight, the right consideration. Right not in moral terms but, so to speak, physical, numerical. Give your parents what is correct, no less, but also no more. What is meant here? The fact that we are all children and that being a child involves many issues, often unresolved throughout existence. Being a child involves living in a tradition, but also in certain expectations, true or presumed. It involves being in certain tensions and disappointments, being heirs of important and significant traditions or disheartening and destructive ones. But less decisive, constraining, stringent than how humans commonly experience them. The past and the roots are sources of deep rifts and wounds, of a desire for fulfillment, imitation, emancipation depending on the case, scenarios from which no one escapes. What remedy does Scripture suggest? Give the right weight. No less, but also no more. You are you, you are not simply the sum, the result, the outcome of those who preceded you. In positive and in negative. You are not the sum of your parents' mistakes nor of their successes and honors. Christ will reaffirm this concept inviting us to emancipate ourselves from the idea that evil comes as punishment for the sins of the fathers. From fathers and mothers, from the past, we certainly inherit a framework of meaning, cultural, technical, of knowledge that each one must freely choose how to manage. How does technique fit into all this? Here lies the point and here I come to the conclusions of this brief intervention.

Technique and technology are an extraordinary tool that allows us to go beyond certain limits of nature, to exist in time and history safeguarding creation and ourselves, possibly in the ecological reciprocity that ensures our planet. Hence the necessity that technique be designed, distributed, and used with an ultimate anthropic purpose, to safeguard human life in its entirety and the balance it must have with life on the planet. Placing humans at the center in the digital metamorphosis can have this meaning. But such an operation must be tempered by taking into consideration the second wound of humanity, its second limit. It would be a fatal illusion to think that the limit given by who we are as persons, as personal history, as negative burden or positive legacy can be, as in biological nature, assisted and tempered by technique. Delegating enabling responsibilities to algorithms, creating parallel worlds in which we do not have to face ourselves and our history, imagining cyborg insertions that allow the body to emancipate itself from its being in tension between an unfinished and an unfulfilled, would be fatal errors. Because they would deprive the human being of its eminent specific: the being in search of itself, exercising freedom in history and simultaneously despite history, being children and at the same time being able to be parents of the future, biological and cultural. The wound of our not being sufficient to ourselves, in other words, must be healed in the relationship with others and with the other, not with the machine that locks us in a virtual solitary state devoid of frictions, but also devoid of personality and thus deprived of our substantial being as persons in search of relationships that fulfill our native existential solitude.

****The technique, ultimately, must allow us to exist in time and in space to concentrate our energies and our capacities in discovering ourselves and the other, delegating to the machine as much as possible what dehumanizes us and retaining for ourselves, indeed asking the machine, to make us**

more capable of exercising responsibility and freedom,
justice and the pursuit of beauty, of
truth, of goodness. With those who live next to us. Those
whom we have chosen, those
who have chosen us, and those who, simply, history places next to us.**

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****Second Session****

AI, Smart City,
IoT, Connected Cars:
Future, Innovation,
Work

**- **MOBILITY AND DIGITALIZATION. SCENARIOS, TRAFFIC,
VEHICLES, PEOPLE****

Prof. Mauro Velardocchia

**- **ARTIFICIAL INTELLIGENCE IN ITALY:
CURRENT STRATEGIES AND FUTURE PROSPECTS****

Prof.ssa Barbara Caputo

**- **NEW COGNITIVE PROSTHESES? DATA IN THE AGE
OF THE INTERNET OF THINGS****

Prof.ssa Anna Maria Mandalari

**- **INNOVATION, AI, AND IoT: REFLECTIONS FROM
PROFESSIONALS****

Dottore Commercialista Paola Zambon

**- **PRIVACY AND DATA MANAGEMENT, VALUE AND RISKS OF A
SYNERGY****

Dott. Marco Gay

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****SECOND SESSION****

AI, SMART CITY, IoT, CONNECTED CARS:
FUTURE, INNOVATION, WORK

****MOBILITY AND DIGITALIZATION.
SCENARIOS, TRAFFIC, VEHICLES, AND PEOPLE****

Prof. Mauro Velardocchia

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****Summary****

Mobility is the result of a multitude of needs and contributions of a technological, legal, and social nature, which have long surpassed the objective defined by the classic perimeter of road traffic and safety. The topic, which falls within the constitutional framework of the freedom of movement, today extends to areas such as transport policies, health, logistics, and the most suitable means to ensure mobility. These are issues subjected to continuous sectoral evolutions, called to interact in favor of improving the many objectives that the term mobility encompasses, addressing complex economic and social questions.

Regarding mobility, the digitalization of information can be considered the common technical language

that allows very different contexts to interact effectively by processing data in a common mathematical format. It thus becomes possible to integrate various capabilities and skills, each influencing mobility, enhancing the peculiarities of each of the areas that contribute to it. One of the most evident practical consequences of such integration of skills manifests in the increase of the level of automation of mobility itself, to respond to very articulated needs in terms of security, traffic flows, emissions, means of transport, territorial planning, logistics, etc.

The development of vehicles presents recognizable evolutionary milestones in relation to technologies. Until the 1970s, the dynamics of a vehicle were entirely dependent on the driver. Subsequently, the first electronic control systems for engines and braking systems (ABS) gave rise to a plurality of driver assistance systems (ADAS). The vehicle, now equipped with sensors, until the end of the 1990s, was still devoid of communication with the outside world, with a focus on functional safety. In the first decade of the 2000s, the vehicle began to communicate with the outside (smartphones, navigation maps).

Georeferencing and sensors (radar, lidar, cameras) were introduced to provide vehicle control systems with information about traffic, people, and signage around the vehicle, thus developing current ADAS. Malicious access to the onboard computer network is detected, with consequences for the dynamic control of the vehicle. The technological focus thus extends to security. Recently, systems capable of detecting the interior of the vehicle have emerged, to observe both the behavior of the driver (incipient drowsiness, state of intoxication, etc.) and that of the passengers. The technological focus begins to necessarily incorporate privacy directives.

A similar technological development has accompanied the digitalization of environmental traffic scenarios (think of maps), the georeferencing of vehicles with increasing precision, the transmission of large quantities of data through radionavigation networks, and the design of road infrastructures. These technological developments are accompanied by the evolution of regulations that govern their use in favor of the collective interest.

Thus, we arrive at the current situation, with a technological scenario in which there is a great abundance of data, obtainable in digital form from the most varied sources: public administrations, communication networks, vehicles, people, infrastructures, etc.

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Digitalization is therefore applied to multiple areas and levels: vehicle, infrastructure, environment, legal, economic, etc. Simplifying, we can essentially identify four areas of digitalization, the first being the context of mobility, in terms of managing signals and data from infrastructures: three-dimensional maps, defining environmental scenarios, acquiring mobility flows, regulations, etc. The second is the realistic reproduction of traffic generated by any vehicle, person, or animal accessing the road infrastructure. The third consists of the description of the characteristics and dynamics of vehicles, more or less equipped with automatic systems, and of drivers. The fourth concerns communication and radionavigation networks. There is an increasing relevance of the capacity to manage large amounts of data (AI) and to integrate a regulatory framework in predictable evolution. Regulations evolve by incorporating the potential that the integration of the technologies summarized above brings with it. For example, EU Regulation 2019/2144 provides for the mandatory equipment of various ADAS (in addition to ABS and ESC, which are already mandatory) and the black box on new passenger vehicles starting from July 6, 2022, for the issuance of EU type approval, from July 7, 2024, for newly registered vehicles, and from 2029 for heavy vehicles. Among the mandatory ADAS systems from 2022 are automatic emergency braking (AEB), lane keeping assistance (Emergency Lane Keeping - ELK), drowsiness monitoring and driver distraction recognition (DMS), assistance for maintaining the most suitable speed (Intelligent Speed Assistance - ISA), the interface for the installation of alcohol lock devices, emergency stop signaling and reverse detection, the event data recorder (so-called black box), etc. This is a technological evolution that has important implications for accidents or in case of violations, a theme of liability for malfunctions. Regarding the increasing implications for privacy, there is an obligation to respect it, ensuring clear management of data in terms of analysis, processing,

storage, flow from onboard devices and communication to the data management centers themselves. It is foreseeable that the evolution of mobility will be progressive and very gradual. What has been indicated previously entails the need for a parallel development of various technologies, skills, regulations, and verification methods that make the entire process safe, advantageous, and sustainable. There is the problem of defining who will bear the cost of this change. Environmental, safety, and economic objectives are nonetheless destined to continue to maintain this trend. This will entail, along with those strictly technological, increasing challenges related to privacy. Finally, one must ask whether we are preparing to consider the individual in the perspective of mobility directed towards increasing automation. If the answer is affirmative, with what methodologies. The presentation introduces for this the use of dynamic simulators as vehicle development systems, sometimes used for the training of professional drivers, capable of verifying how the driver will interact with any traffic scenario, communication network, vehicle setup, weather condition, etc., in conditions of safety and repeatability of any driving situation. It is a methodology based on the digitalization of what contributes to mobility, which allows for the preliminary analysis of the effects introduced by the evolution of any of the presented elements. The analysis of the data flow that allows such dynamic simulation systems to safely reproduce what may actually occur in a mobility scenario is thought to enable the preliminary verification in a laboratory environment of the methods of data processing themselves in order to confirm compliance with privacy regulations.

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ARTIFICIAL INTELLIGENCE IN ITALY:

CURRENT STRATEGIES AND FUTURE PROSPECTS

Barbara Caputo

Professor of Artificial Intelligence, Polytechnic

University of Turin

Artificial Intelligence is undoubtedly the science whose technological repercussions will have the most disruptive transformative impact on the economy, society, security, defense, and geopolitical balances globally in the twenty-first century. A technology with such disruptive effects must be monitored at various levels: scientific, technical, legislative, and strategic. For this reason, all industrialized countries (and not only) have dedicated significant attention to the topic over the last five years, equipping themselves with National Strategies to plan investments in support of development policies. Similarly, various international actors have promoted the creation of associations and permanent forums on Artificial Intelligence, to seek to

build a common language for dialogue on this topic, and identify shared principles to govern its development. Italy is no exception, and in 2021 the country adopted a Strategic Program for Artificial Intelligence 2022-2024. The document,

resulting from the collaboration of the Ministry of University and Research, the Ministry of Economic Development, and the Minister for Technological Innovation and Digital Transition, identifies twenty-four strategic policies for enhancing the Artificial Intelligence system in Italy.

The overall vision is, in line with the European Strategy on the subject, that of an AI that places humans at the center, identifying education, fundamental research, and innovation as the three fundamental pillars around which to structure the 24 lines of intervention. Some of the actions identified by the strategy, such as the establishment of a national PhD in artificial intelligence and the creation of a widespread research ecosystem sponsored by PNRR funds, are underway, while others are about to start, such as the AI center for automotive and aerospace, which is expected to act as a catalyst for innovation in these two strategic sectors for the national economy.

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It is important to be aware that the current Strategic Program was conceived during the height of the pandemic emergency, envisioning a post-COVID future of significant economic recovery on a global scale.

That imagined future has not materialized.

We now find ourselves in a different world, with radically different geopolitical balances and a new perspective for the future. It is essential to start working now on the new Strategic Program for Artificial Intelligence 2025-2027, a program that takes into account this changed global context and is developed by involving the academic world as well as the productive sector, the top levels of security and defense, and the Public Administration as a whole.

It is also crucial to initiate a reflection on the opportunity to recognize Artificial Intelligence the strategic importance that has been granted to Cybersecurity and Aerospace, a recognition that leads to assigning to Artificial Intelligence the powers of direction, coordination, programming, and oversight to the Presidency of the Council of Ministers.

Recent news events, with the global launch of ChatGPT by OpenAI, the unrestrained and interested emphasis regarding its capabilities to automatically generate texts and respond to requests across a wide spectrum

from users in a more or less competent manner; the emergence of issues regarding the truthfulness and reliability of the content of such texts and the reaction of the Data Protection Authority to defend the rights of citizens and minors, illustrate how the Artificial Intelligence landscape is complex and only at its beginning.

The hope is that Italy can respond promptly to this challenge and become a reference point and critical spur at the global and European level, where a regulatory proposal in the form of the AI Act has long been awaited.

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NEW COGNITIVE PROSTHESES?
DATA IN THE AGE OF THE INTERNET OF THINGS
Prof. Anna Maria Mandalari
Professor at the Institute for Security
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We know that any object today can be connected to the Internet, from smart speakers to security cameras; even the refrigerator can be connected to the Internet. We call these devices the Internet of Things (IoT).

The Internet of Things is a very powerful tool and allows us to live a comfortable life.

But why do they cost so little and what is the real value they provide us?

This is where my research journey began; I wanted to understand what we are invisibly offering in exchange for these devices.

In this context, we must consider three fundamental steps:

- 1) Privacy versus minor expedients.
- 2) How IoT devices expose privacy.
in a smart home
- 3) What is the impact that devices will have in the future.

At the beginning of this journey, we studied how IoT devices handle user data and explored in depth whether they violate user privacy. The main motivation for this research is that these devices are considered "black holes"; they can (by definition) access the Internet and therefore can expose private information.

Moreover, there is a lack of understanding about what information they expose, when they expose it, and to whom. The regional differences in regulations are not clear (e.g., GDPR is present in Europe but not available abroad).

Our initial research includes the following points:

- 1) What destinations are contacted by IoT devices?

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- 2) What information is sent?

3) Do they behave unexpectedly?

To answer these questions, we purchased over 200 different devices, from appliances to smart cameras, smart hubs, and other home automation devices, and placed them at Imperial College London and in a studio apartment at Northeastern University (NEU). The IoT lab at Imperial College is managed by me and Professor Hamed Haddadi, while the one at NEU is maintained by Italian researcher Daniel Dubois and Professor David Choffnes.

Our research methodology is based on collecting the Internet traffic produced by these devices and conducting automated large-scale repeatable experiments. Figure 1 shows the labs in the UK and the United States.

Figure 1: IoT Labs in the UK and the United States.

Data collection occurs on the router, which distinguishes traffic by device and by experiments; thus, the behavior of the device is completely isolated.

As shown in Figure 2, from an initial investigation, we found that most of the traffic from the UK lab goes to the United States or China.

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Figure 2: Internet traffic diagram of devices.

Traffic from many categories of IoT devices is sent outside the privacy jurisdictions of our lab.

We found other privacy violations, for example:

- 1) Some smart doorbells recorded at every movement and sent the recording to a cloud server in another country even though this had been disabled in the app.
- 2) Smart TVs contacted Facebook, Netflix, Google, and some advertising trackers.
- 3) Smart speakers activated without saying the wake word.

We then studied the behavior of smart speakers more closely to understand if they activated without the user's permission.

From our experiments, it emerged that smart speakers often activate unexpectedly, with keywords similar to the activation word, even activating incorrectly for up to 10 seconds.

We also discovered that most of the traffic is not sent to the manufacturer but to "third parties," such as trackers and advertisers. Often, the contract established with the manufacturer allows them to transfer your data to third parties, but the user is unaware of this, as the terms and conditions are too vague. This creates a risk to privacy.

The main issues are multiple:

- 1) Profiling - Based on the devices and the traffic from the devices. Third parties can create a unique profile of the user. Preferences, interests, it is easy to understand what the user likes and influence future choices.
- 2) The greatest risk is the so-called "mass influence."
- 3) Voice signals are also a rich resource for third companies, as they reveal various possible states of a user, such as emotional state, stress levels, physical condition, age, gender, and personal traits. Service providers can create a very accurate profile of a user's demographic category, personal preferences, and this can compromise privacy.

When we think about the future of these devices, we know they will be much more sophisticated. What will happen when these devices become more like cognitive prosthetics? Helping the user make decisions on the most intimate topics. With access to user data, technology platforms will learn preferences, anticipate needs and behaviors, and monitor health. But in reality, in this future, platforms may emerge that use user data for the wrong reasons.

In my vision of the future of IoT devices, control will again be in the hands of the consumer. The user will be aware of when information is used negatively or for unclear purposes. All IoT devices in use will be certified and reliable with solutions no longer present in the cloud but close to users, at the edge.

I producers must declare the destinations that the devices are contacting and the reasons for contacting specific destinations (particularly third parties). The certificates will include security, privacy, and energy consumption. Figure 3 shows a prototype of an IoT certificate.

Figure 3: IoT Certificate Prototype.

At Imperial, we are also developing software that allows for the protection of user privacy and security. From our analysis, it emerged that part of the IoT traffic is essential for the operation of the device, while part is non-essential. We have therefore developed software that allows for the "blocking" of certain non-essential communications while continuing to use the device perfectly. This software, functioning in the home router, is named IoTrim (Figure 4).

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Figure 4: IoTrim.

IoTrim, using artificial intelligence techniques, automatically identifies all devices connected to users' routers and non-essential traffic, thus protecting user privacy and security and keeping IoT connections private and secure.

INNOVATION, AI & IoT: REFLECTIONS FROM PROFESSIONALS

Chartered Accountant Paola Zambon

Joint GDPR Table of the Order of Chartered Accountants and Accounting Experts, Lawyers, and Engineers of Turin

In the session dedicated to IoT, AI, and innovation, Paola Zambon spoke on behalf of the Joint GDPR Table of the Professional Orders of Turin (Chartered Accountants, Lawyers, Engineers), providing reflections on AI and IoT from the professionals of the Table itself. The speaker began by thanking the present authorities and conveying the greetings of the presidents of the professional orders, and then recalled that the Table itself was the first in Italy to provide support to professionals and businesses in operational clarifications of the GDPR in line with the best interpretations of business law and always following the guidelines of the Guarantor Authority.

Technology should always be placed at the service of people in full safety and in respect of the values and fundamental rights of the EU, even according to the most recent European guidance, despite the increasingly close interfacing between man-machine-environment.

The anthropocentric and sustainable digital transformation, therefore, is what businesses aim to achieve with the added value that artificial intelligence systems can bring, some even with the use of big data, but there are still paths to be taken on certain issues (e.g., the lack of adequate skills in the market (both technical and professional), the dangers of social manipulation and mass surveillance). The principle of accountability must concern both businesses and individuals, who are increasingly informed about the risks arising from the processing of their personal data but often still unaware of the effects that the same can have in their lives, both due to underestimation and the opacity of the models present in the market.

Business law has been increasingly permeated for years by intelligent agents and artificial intelligence systems, in which the European model aims to be anthropocentric, ethical, fair, secure, reliable, and sustainable.

In this perspective, the professional highlights that the macroeconomic framework of Italian companies does not enjoy an excellent international positioning as massive private investments are mainly made from abroad (e.g., USA, China). The circulation of data occurs and has occurred mainly through the use of IoT, as due to the pandemic, people have changed habits and lifestyles, and there has been an exponential increase in the use of non-European platforms for many types of services. It cannot be

overlooked that the "declaration on digital rights" at the European level establishes an important milestone, complementing existing principles and considered an attempt at European innovation in search of a compromise between different international models in terms of culture and different levels of technological evolution. The new "virtual worlds" must incorporate European values from the outset so that people can feel protected as if they were in the real world.

The very definition of artificial intelligence is much debated and of strategic importance in the market. In the search for "standards" for this topic to be implemented with particular attention to humans, the involvement of professionals who have gained operational experience on these issues, even alongside SMEs (which often struggle to enter such markets), is deemed essential.

For the purposes of the intervention, therefore, after illustrating the undeniable advantages that technology and AI can bring (e.g., in scientific research related to health, agriculture, the environment, etc.), attention was focused on some areas of reflection that can also impact the correct governance of companies, increasingly connected to such systems, although not always accompanied by adequate knowledge of the GDPR and, in general, of the applicable regulations. These macro-themes are:

1. Correct governance for companies that implement/adopt AI systems
2. Ethical design and privacy by default and algorithmic transparency
3. Non-discrimination in AI systems and the "embedding" of a culture of diversity and inclusion
4. Sustainability.

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Correct governance for companies that implement/adopt AI systems

What can be considered essential in a company that produces or adopts artificial intelligence systems
Some issues to reflect on

- Principles of Privacy by Design and by Default
- Right to information for data subjects and application of the principle of transparency
- Human intervention in automated decisions
- User awareness of profiling information obtained from cross-referenced metadata (emotional reactions, cognitive abilities, mental health, preferences, tastes of all kinds, consumption, etc.)
- Guaranteed service (to avoid the risk of unavailability of data, which could be very dangerous)
- Accountability
- Application of quality guarantees and ethical design
- Issues related to skills: often there is a lack of adequate competencies and profiles
- Issues related to security: an attack on IoB can seriously jeopardize people's health, even their lives, mass surveillance
- Issues related to sociality: social manipulation, modification and influence on human behavior, etc.
- Issues related to competition: control over IoT data could create a risk of vendor lock-in and user dependence on a few producers
- Issues related to environmental sustainability: the energy consumption required by these services leads to an increase in CO2
- Issues related to inclusion: often women lack skills in AI, algorithmic discrimination, misinformation, etc.

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The "ethical design" concerning companies governed by or adopting artificial intelligence systems should pertain not only to the well-known principles of privacy by design and by default but also encompass a realistic implementation model of ethics, so that it proves reliable. Organizational models in this regard often report ethical codes that do not work for various reasons: lack of resources to make them function, opacity in the decision-making process, lack of independence from the governing body, and often ignorance of the applicable regulations with the presence (if not absence) of committees that, if they function, are not listened to. In this sense, as a Table both as business jurists (accountants and lawyers) and as technicians (engineers), the added value we can bring to the market is significant,

even at the level of responsible awareness. Furthermore, the focus is emphasized on the importance of the correct way to balance the method with which the training conducted impacts decision-making complexity and algorithmic transparency (also to avoid GiGo-like approaches, which would also be contrary to the principles of social responsibility) typically protected by trade secret with the opacity of decision-making models found on the market, also avoiding the use of dark patterns that deliberately make it difficult for users to understand the risks they may incur.

Allogocracy should not lead to misinformation: the role that professionals can play in signaling inappropriate technical content, especially to the detriment of companies, is important. Another major challenge is represented by the right to gender equality and, in general, the prohibition of discrimination, which is often found violated in the application of AI. "Algorithms and the associated machine learning, if not sufficiently transparent and robust, risk reproducing, amplifying, or contributing to gender biases of which programmers may be unaware or that result from a specific selection of data." For example, studies conducted on clinical and health samples usually rely on white male subjects (thus it is likely that they are not suitable for other genders and people of different colors).

This has also emerged from the experience of the ICT Association.

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Dott.Com, to whose events the Table participates, the gender bias in the algorithmic decision-making process is not only manifested in the lack of adoption of correct evaluations to be made at the design level, but is also found in the market due to the lack of diversity among the professionals who design, program, engineer, and maintain artificial intelligence technologies. It is also noted that there is an objective difficulty both in terms of training and monitoring processes related to AI systems, such as established criteria in the measurement of multimodal "bias" or the difficulty of exercising the right to be forgotten in machine learning. In many technology companies, leadership is solely male, and there is neither knowledge nor competence regarding diversity and inclusion.

Some intelligent agents also have a significant social impact, and the risk of "gender" social surveillance is always looming. For example, in the USA, some women have uninstalled menstrual cycle apps to avoid being tracked when seeking an abortion (in some states, abortion is considered illegal). Some companies have thus decided to delete geolocation history when a person visits an abortion clinic, a domestic violence shelter, or similar places.

Finally, from a sustainability perspective, it is deemed useful to also understand the environmental and social impact, including in relation to the changing profiles and methods of work that the artificial intelligence system adopted by a company brings to the community, and thus also evaluate the benefits it produces concerning the expected energy consumption for these services, which typically lead to an increase in CO₂ (climate and environmental risk).

In this context, accountants are already accustomed to dealing with non-financial information related to companies that, under the principle of business continuity, will increasingly be directed towards ESG objectives, within the broader framework of corporate social responsibility, which has so far been largely left to the free discretion of entrepreneurs, but which, with the incessant proliferation of regulations impacting the digital transformation of the company itself, is becoming increasingly complex. Information concerning intellectual, human, social, and relational capital will also be important.

It is also important to involve professionals in the adoption of technical standards, as the presence of cross-disciplinary skills is deemed essential to highlight certain aspects that are usually not evaluated by technicians, such as:

- Data quality and accountability: it should not only be associated with mere technical standards proposed in the market but also with sustainability and inclusion, taking into account the non-financial informational impact. The data scientist (rare to find in the market) needs to be supported by professional figures who can transfer other skills to him/her, including in the construction of the "data policy";

- Preservation: this aspect is usually widely underestimated even in terms of technical regulatory compliance (not only concerning privacy regulations);
- Standards yes, but with caution: there is a risk of the proliferation of too many standard proposals and too many interpreters in the market: they should be carefully evaluated. While some are useful, others could become de facto rigid regulatory tools, disregarding the very concept of "accountability," which could lead to the emergence of new cases before the EU Court of Justice. It is important to understand who will actually evaluate the compliance of certain systems, especially those related to AI. Finally, data can provide added value if constructed by adequate teams (not only technical but also professionals with experience in data management and privacy).

Finally, recalling how professionals, particularly accountants, lawyers, and engineers from Turin, are available and ready to assist SMEs in digital transformation that is human-centered and sustainable, reference is made to the conference on November 18, 2022, at the Polytechnic of Turin organized as always by the ICT Association Dott.Com, which was attended by the Data Protection Authority (Dr. Agostino Ghiglia), an event in which some of these concepts were also revisited.

CONFERENCE ON THE OCCASION OF THE CELEBRATIONS FOR THE 25TH ANNIVERSARY OF THE DATA PROTECTION AUTHORITY - JULY 4, 2022 - TURIN PRIVACY AND DATA MANAGEMENT, VALUE AND RISKS OF A SYNERGY

Dr. Marco Gay

President of Confindustria Piemonte and ANITEC-ASSINFORM

Dear colleagues, esteemed speakers,
it is a true pleasure to be here with you today.

The 25th anniversary

of the establishment of the Data Protection Authority gives us the opportunity to reflect and discuss the future that data will have in the economy and in society. What I would like to focus on is the role that businesses will have in protecting and enhancing them.

The digital transition pervasively involves society and the economy. From the city, to the factory, to the home, products and services are increasingly advanced, versatile, and efficient. Compared to just a few years ago, the amount of data we produce is

impressive. This is especially due to IoT and connected devices, technologies that are widespread both in the industrial sector and in everyday life.

Not only has the volume of data increased, but so has our ability to leverage it. New technologies have laid the groundwork for the birth and development of the so-called "Data Economy." Leveraging data, in an increasingly refined and effective manner, goes far beyond their monetization; we are dealing with the broader concept of value creation. An adverse consequence of the ability to create value from data is the increase in cyber risk.

For this reason, we must multiply efforts to protect the data we

hold.

In this context, innovation and regulation are two indispensable elements that, despite being in tension, need to support each other.

The European Union has long begun the construction of its

“digital constitution.” It started with the GDPR, the Regulation on personal data

which has established itself as a global standard for data processing, and has

continued to ambitious proposals such as the Data Act and the AI Act.

The AI Regulation represents the first attempt to regulate

Artificial Intelligence at a global level.

The goal is - on one hand - to

stimulate innovation in the EU and - on the other -

to create a reliable and credible development environment for citizens. An AI ecosystem

that puts “humans at the center.”

It is undeniable that the European Union has chosen to act on

regulation to participate in the race started by

the USA and China, two global superpowers

that are now far ahead in research,

investments, and patents.

At the same time, there remains a significant risk of inhibiting innovation, and

in particular of overwhelming SMEs and innovative Start-Ups

with regulatory costs and

bureaucracy. As always, when it comes to

regulation and innovation,

for a policy to work, mere regulation is not enough and is indeed harmful.

In fact, the most promising tools for

privacy protection and

for the security of our online lives come from

scientific research and

cutting-edge technological applications, such as advanced

AI solutions

in the field of Cybersecurity, Edge computing, and synthetic data.

- AI is increasingly being adopted in Cybersecurity.

We know well

how this area is now central to

the activities of any

business, even the least digitized. In cases of

advanced attacks, AI solutions can help identify

anomalies that

raise suspicion of an attack and allow for

timely detection.

- In the field of data processing and computing,

the paradigm of Edge Computing is emerging, a technology

that ensures

significant advantages in terms of data protection.

Information remains within the company perimeter, does not require sending to centralized areas for processing but can be processed directly near the user.

- Synthetic data is certainly a niche and cutting-edge solution but has a rapidly growing market. Companies use them not only to train more precise and performant AI algorithms but also to improve compliance with privacy regulations. There is a vast difference between using sensitive data, such as biometric data from real people, and synthetic biometric data generated by a neural network.

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In conclusion, the security of our digital lives does not rely solely on regulation, which must exist but should have defined contours that do not hinder entrepreneurial freedom, but also, and above all, on innovation.

In this sense, the ICT industry has a social function, in fact.

Being enablers of progress means committing to ensuring that all citizens have access to the most cutting-edge technologies and the highest security standards in all products.

THIRD SESSION

the authority
of the future

- SPEECH BY Prof. Pasquale Stanzione
President of the Data Protection Authority

- DIGITAL HEALTH AND PERSONAL DATA PROTECTION:
A POSSIBLE COMBINATION

Prof. Ginevra Cerrina Feroni
Vice President of the Data Protection Authority

- METAVERSE, PRIVACY, AND THE FUTURE
Prof. Guido Scorza
Member of the Data Protection Authority

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****THIRD SESSION****

****THE GUARANTOR OF THE FUTURE****

Prof. Pasquale Stanzione
President of the Authority for the Protection of Personal Data

Today we discussed the 25th anniversary of the Guarantor.

A common element in the various reports is the awareness of how fundamental these 25 years have been in outlining a governance of sustainable innovation in democratic terms. The very concept of data protection has undergone a significant evolution, from the right to be left alone to the right of

informational self-determination, which is expressed in the governance over the circulation of our personal data. The transition from privacy, traditionally understood as a prohibition against interference in the private sphere, to the right to data protection has been somewhat of a common thread underlying the debate. The value of personal data has been emphasized as fragments of our self, divided (in the fragmentation caused by the network) into many micro-identities, referring to a part (and only a part) of us, which represents us but not entirely: voter, consumer, minor, prisoner, patient, victim, and so on. The challenge today is to restore wholeness to the fragmented person and place them at the center of technological evolution. It is no coincidence that during the presentation of the annual report, we will talk, on July 7, about digital humanism, that is, the need to functionalize innovation (which would otherwise be paradoxically regressive) for the protection of the person. A prerequisite for this goal is the conscious governance of technology by humans, who must not merely endure but guide innovation, which in this sense is a common good. And it is a common good to the extent that it enhances rights and freedoms and does not legitimize or, worse, amplify violations (as occurred with the discriminatory consequences of the use of certain algorithms).

* Dr. Federico Monga, Deputy Director of La Stampa, interviewed the members of the Guarantor's College.

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Determinative, for this purpose, are the principles of transparency, non-discrimination, and the right to an explanation of algorithmic decisions enshrined in the GDPR.

Thus, drawing conclusions from the various interventions, we can say that progress must be promoted but with an attitude distant from both uncritical scientism and technological solutionism, as well as from neo-Luddism. Innovation therefore requires a true digital humanism that values the personalism on which our Constitution is based.

****Roots and Horizons of Data Protection****

Reflecting on the evolution of data protection law implies, at the same time, a reflection on power and the rule of law today, on the ability of the latter to ensure a governance of innovation that reflects its axiological system, thus complementing the rule of law with an equally unavoidable rule of technology. Just as modern constitutionalism has established itself through the progressive construction of limits on state power in function of individual guarantees, so today the process of governing innovation cannot disregard the definition of some essential rules for the protection of the person, in relation to the new power structure induced by digital technology.

To fully understand the implications of the performative power of technology on the person, we can take as paradigmatic some essential themes: identity and its representation online; the construction of personality and the formation of public opinion; the maintenance of individual guarantees in the face of algocracy.

The first aspect allows us to highlight how the digital realm has dismantled not only the traditional system of power allocation but also the process of identity construction and, therefore, its relationship with freedom.

If the term 'identity' is a singularia tantum, it is because it has never been conceived in the plural, represented as it has been by tendentially immutable coordinates among which is the name. New technologies, however, have made the term "identity" necessarily plural, alongside the physical identity, a kaleidoscope of digital identities that compete, almost to the point of prevailing, over the former (the user, the voter, the consumer, etc.).

In this context, data protection has played a central role in the "recomposition of the divided self," fragmented into a thousand pieces scattered across the network, ensuring the integral representation of the person. This is a result to which the judicial development of the right to be forgotten has also contributed, as a tool for protection against the crystallization of the self in often distorting details. However, the performative power of technology over the person is not limited to their representation but extends to touch that "inner world," that internal forum considered an inviolable bastion of intangibility as an essential prerequisite for self-determination: the free formation of individual

conscience. The hierarchy and selection of news determined by algorithms, which show or highlight only certain content and not others, are a paradigmatic example of how new technologies condition the very formative process of public opinion.

Paradoxically, a society such as the digital one, potentially capable of guaranteeing maximum information, risks instead artificially limiting access to news, fostering conformity and undermining informational self-determination. Information

risks thus degenerating into “mass self-communication” (Manuel Castells) and political nudging, made possible by propaganda tailored to the profile of the voter attributed to the user by the algorithm, as in the case of Cambridge Analytica, risks deconstructing democratic dynamics from within. Even with respect to this type of distortion of the process of forming individual consciousness - with significant implications for public life - data protection law offers some tools, especially preventive ones, for protection. Foremost among these is the provision of an essential limit to algorithmic control, partially entrusted to individual will through the consensual device and, partially, to the heteronomous assessment of the non-abusiveness of the commercial exploitation of individual data. Indeed, they are the primary resource of the new private powers, protagonists of that form of extractive capitalism characteristic of the platform economy.

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The guarantees provided by privacy law are more than necessary with respect to what can represent, if not adequately modulated, a combination capable of exacerbating inequalities: the functionality of technology in consolidating asymmetric social relations and in the exercise of power, detached from an adequate system of accountability. The suspension of the social media accounts of then-President Donald Trump symbolizes, in a paradigmatic way, the impact of the private power of platforms on the guarantee of a fundamental right such as freedom of expression. Regarding the risk of relegating the perimeter of exercising freedoms online to contractual clauses, data protection law establishes some essential principles aimed at countering the abuse of private power by platforms. These principles are enhanced and complemented by those newly introduced with the Digital Services and Digital Markets Acts, aimed at the overall accountability of platforms in terms of (especially) protecting the individual.

However, an anthropocentric governance of innovation is necessary, especially concerning the increasingly frequent use of algorithms in public policies. Recent experience demonstrates how technology has become an unavoidable tool for governing emergencies, social complexity, and emerging needs. It is a tool that can be all the more valuable if well governed, but also potentially disruptive if lacking the necessary limits. This is especially true (but not only) for the use of AI in law enforcement activities, where computational power is instrumental to the monopoly of legitimate violence. These considerations are also underpinned by the stringent limits imposed by the Artificial Intelligence Act on facial recognition in public places by law enforcement authorities, which is particularly prone to degenerating into forms of mass surveillance. However, data protection law already provides important safeguards for the use of AI in law enforcement activities, which have led the Guarantor to deem a facial recognition system for public safety purposes inadmissible, lacking the necessary guarantees to bring otherwise massive measures in line with proportionality.

In this sense, the moratorium on the use of facial recognition approved during the conversion of decree law 139 is significant, as it does not abstractly and entirely preclude facial recognition by law enforcement authorities but conditions its admissibility on the favorable opinion of the Guarantor, ensuring that any possible use always incorporates the necessary privacy guarantees. Indeed, today, data protection law provides some essential regulatory criteria for the development of AI consistent with the values of the European legal order and constitutional principles: foremost, the primacy of the individual and their dignity, which is even more relevant in light of the injunctive turn of technology, increasingly demiurgic, predictive, and therefore performative. In the face of such revolutionary (but also potentially socially regressive) innovation, which could almost invert the instrument-agent relationship that should guide its interactions with humans, the legal status of AI outlined by data protection law is crucial to truly place technology “at the service of man.” Consider the principles of

non-exclusivity of automated decision-making (which prohibits, barring the existence of adequate guarantees, the complete delegation to the algorithm of significant decisions about individuals), non-discrimination, and comprehensibility (and thus also reviewability) of algorithmic assessments, aimed at countering the risk of bias.

In this strategy of "humanization" and functionalization of technology to freedom, data protection law is destined to play a primary role, even beyond AI: as a limit and as a horizon of meaning. Let us all contribute, through respect for its norms and awareness of its relevance, to promoting its value.

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DIGITAL HEALTH AND PERSONAL DATA PROTECTION:
A POSSIBLE COMBINATION

Prof. Ginevra Cerrina Feroni

Vice President of the Italian Data Protection Authority

The combination of digital health and personal data protection is possible. This is demonstrated by the numerous interventions of the Authority aimed at facilitating a balanced and sustainable development of healthcare services offered to citizens, taking into account the protection of fundamental rights of individuals in light of social progress and scientific and technological developments. Positive examples of this balance are evidenced by the information systems and technological solutions introduced in the emergency context for the management of vaccinations (Information Platform for the implementation of the strategic plan for anti-Sars Cov 2 vaccines) and Covid-19 green certifications (National Digital Green Certificate Platform - PNDGC), on which the Authority has provided its opinions (respectively on January 13, 2021, and June 9, 2021). These are challenges overcome, demonstrating that an effective healthcare information system, which allows for immediate availability of correct and updated information, can be realized in ways that ensure data protection and the rights associated with it.

In this regard, the Authority has shown its willingness to support the Administrations engaged in the implementation of significant reorganization initiatives such as those arising from the implementation of the PNRR projects. Obviously, this is done in respect of the independence of its role in overseeing the implementation of European and domestic legislation on personal data protection. A mission, not a whim - it is useful to reiterate - of service to citizens and safeguarding their rights and personal freedoms that could be profoundly impacted by a distorted use of personal information, particularly that of a health-related nature. In the opinions issued on August 22, 2022, regarding the decrees related to the Electronic Health Record (FSE) and the Health Data Ecosystem (EDS), for example, many critical elements were identified, most of which stemmed from the fact that these are systems whose functions and possible uses are still not fully understood.

As is known, European data protection legislation stipulates that the legal basis for processing must indicate which data to process, how to process it, who can process it, and specify the technical and organizational measures to be followed. From this perspective, the decrees on which the Authority expressed its opinion should have implemented a clear project, within which the privacy impact should also be considered through an adequate "risk assessment." Instead, they were found to lack such elements, containing general and vague formulations that would have generated confusion precisely in those sectors - such as healthcare - where clarity of procedures is a fundamental element for patient safety and medical professional responsibility. Allowing access to incomplete information, for which the update and constant availability of the data are not guaranteed, is not only a critical element for data protection but also for the quality of care provided, especially in emergencies.

The decree on the EDS was, for example, evaluated by the Authority as a "empty box" in which services for doctors, personalized for individual patients, are announced, but without indicating what these services are and on what personal data they are based. Similar considerations were expressed regarding the FSE: the decree should have contained, as required by law, concrete indications on the use of the record, but there are still many blank spaces that leave doubts and uncertainties even among healthcare operators. Therefore, the Authority could only refer the projects back to the

Government, requesting the necessary clarifications regarding the lack of essential elements that, instead, were ensured even during the most intense moments of the pandemic, namely project planning and sharing.

It is worth noting that the issue of health data has taken on a supranational dimension, given that on May 3, the Commission published the proposal for a Regulation (of the European Parliament and of the Council) on the European Health Data Space, the so-called EHDS. In summary, this proposal aims to strengthen the control of the

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****Personal Data Protection: Granting New Rights to Data Subjects Regarding Their Electronic Health Data****

The aim is to facilitate the use of such data by fully leveraging the potential offered by the circulation, exchange, and secure and protected reuse of it. This is intended both to provide better healthcare, including cross-border care, for the individual to whom the data pertains ("primary use"), and to support the formulation of health policies, facilitate scientific research, promote the development of personalized medicine, produce official statistics, as well as foster innovation in general in the health sector, including the development of algorithms, AI systems, and digital health apps ("secondary use").

However, the use of electronic health data for a series of additional purposes is prohibited, such as making decisions detrimental to individuals or aimed at excluding individuals from insurance benefits or increasing insurance premiums, commercial advertising, and the development of services and products that are dangerous to individuals and society in general.

The proposal, however, raises a number of general concerns regarding aspects related to the protection of personal data, which have been highlighted by the European Data Protection Board (EDPB) and the European Data Protection Supervisor (EDPS) in a detailed joint opinion (3/2022), adopted on July 12, 2022, in which greater clarity was called for regarding the concrete possible uses of health data in this sharing space and a better delineation and circumscription of possible activities. In particular, attention has focused on Chapter IV of the proposal, which aims to facilitate the secondary use of electronic health data. The reuse of health data from European citizens, while potentially generating benefits for the public good, cannot be considered free of risks to the rights and freedoms of individuals, specifically regarding the protection of personal data.

And therefore? What are the prospects for digital health in terms of personal data protection?

Let us start from the consideration that privacy and health share a common language. Many of the measures requested and obtained by the Authority in recent years have concerned the introduction of tools that allow, while facilitating the use of technology in healthcare, to ensure the quality and accuracy of the data to which healthcare professionals have access. Obtaining health information in real-time is undoubtedly an advantage in the care pathway of a patient, but the speed of connection must not come at the expense of the correctness of the data and certainty regarding its update and integrity. These are elements on which the Authority pays great attention, as demonstrated by the opinions on the Transplant Information System (March 30, 2017) or on Advance Treatment Directives (May 29, 2019).

When the Authority expresses its opinion on legislative proposals that aim to innovate health information systems, it does not limit itself to verifying the robustness of IT measures, the security of systems and platforms, but primarily evaluates the correctness of the processing and the protection of the fundamental rights of the patient, which also involves accurate, updated, and unaltered data processing. In this sense, contrary to what some, even distinguished, misinformed individuals may think, data protection is not an obstacle or a brake on innovation and technological progress in healthcare, but can indeed be an accelerator and a virtuous guide.

If anything, the crucial issue is that the problems of digital health in Italy are structural.

Let us stick to the data. The FSE system is formally implemented in all Regions completely or almost, but essentially remains on paper. With the exception of Emilia Romagna, Lombardy, and Lazio, the average of patients who have given their consent to the feeding of the FSE stagnates around 6%, with as many as 10 Regions where no patient has given consent. As for doctors, excluding Sicily, Umbria, and Valle d'Aosta, in the other Regions, 0% contribute to the health profile of patients, and in as many as 8 Regions, no doctor or almost consults the FSE.

Then there is the issue of training and information, on which there is much to be done. What is lacking, in order to benefit from all this innovative and complex technological arsenal available, is adequate education for users on all its potential and the methods of use and access.

A logistical reorganization of public health, a renewal of its infrastructures, the provision of interoperability of systems, as well as a profound training effort for IT and network professionals, are necessary.

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****Personnel at all levels appear as essential elements in order to facilitate a correct positioning of the pieces that make up digital medicine and the potential that this tool can express, but which, instead, too often remain crushed by existing infrastructural and budgetary constraints.****

The opportunity to partially mitigate the delays of the country regarding digitalization and the use of information systems, and consequently to hypothesize a more structural reform of both public and private healthcare with a strong push towards telemedicine, must be the PNRR (National Recovery and Resilience Plan), in which over 40 billion are earmarked for digitalization and a good part of the 18 billion for healthcare will be available for home care and related processes.

However, we must remember that when we talk about digital healthcare, the purpose of care does not change. What changes, more or less directly and from different perspectives, is everything that surrounds it: the ways in which medical services are delivered, the relationship between those who provide and those who benefit from them, and the organization and delivery of related services, with the consequence of strongly emphasizing the happy expression that taking care of the patient means also taking care of their data, no longer just the physical body as Hippocrates and Galen did, but also the digital one.

1) Cf. R. Balduzzi, Some conclusions: the difficult equivalence of regional healthcare subsystems, in E. Catelani, G. Cerrina Feroni, M.C. Grisolia, Right to health between uniformity and differentiation: models of healthcare organization compared, Giappichelli, Turin 2011, pp. 149 et seq.

****DATA PROTECTION. FOR 25 YEARS THE GUIDING STAR OF THE FUTURE****

****METAVERSE, PRIVACY AND FUTURE****

Prof. Guido Scorza

Member of the Authority for the Protection of Personal Data

In questioning the Metaverse, privacy, and the future, I believe it is essential to first turn our gaze to the past, to the history of the Internet and the governance exercises of the technological phenomenon that has changed our lives, the society we live in, the markets, politics, and the world more than any other. The Internet was born in the public dimension, within a military project, to be precise. It was called, as many of us remember, Arpanet and was supposed to ensure communication even in the event of a military attack.

It was literally privatized and then landed in the world of universities and research, and only later was it transformed, essentially, into a market tool in which private powers increasingly erode public powers, appropriating roles, competencies, and functions that should be the exclusive domain of the latter.

We do not know what the Metaverse will be, and, indeed, at this stage, we do not even know if it will be necessary to speak of it in the singular or plural.

However, there is one certainty in investigating its origins, and that is offered to us by Google Trends, the Google service that monitors user queries on the search engine: essentially no one, in the whole world, had queried the search engine using the term "metaverse" until October 28, 2021, when Mark

Zuckerberg, founder and head of Facebook, announced that his company would change its name to Meta because the world was heading towards the "Metaverse."

From that moment on, the graph of searches on Google using the word "metaverse" suggests that hundreds of millions of people around the world began to take an interest in a future, baptized Metaverse by the number one of a company that had just chosen to call itself Meta.

If there is, therefore, a reasonable certainty regarding the Metaverse that will come, it is that it will be a more private non-place than the Internet from its very first moments of life.

In short, unlike what originally happened with the Internet, to access the Metaverse - or, if there are to be different ones, the metaverses - we will necessarily have to accept general terms of contract that will establish, just as happens today when we start using a social network or another digital service, what we can and cannot do in the metaverse or metaverses.

The metaverse will, therefore, be an enormous private garden whose owner will dictate the rules that billions of users of that garden will have to respect, and the rules in question, unlike the laws enacted by Governments and Parliaments, will be rules that cannot be violated since algorithms and software will oversee compliance by all, simply preventing any future violations.

This is a first consideration that cannot be overlooked when addressing the legal issues related to life in the metaverse because, in essence, it suggests that the scope for public powers will be even more limited than it is today in the digital dimension and that the scope for private powers will be, essentially, unlimited.

In short, we will be free by contract or, perhaps...

worse, we will not be free by contract because, indeed, the contract in question will significantly compress our rights and freedoms.

A significant problem to manage if we want to avert the risk that what could be destined to become, in a modest temporal universe, the natural living space of man is almost completely removed from the rules, governance, and justice that have so far shaped our democracies.

And, the problem within the problem, is that, presumably, none of us will actually know the rules to be followed to live in the metaverse because, just as is the case today with terms of use and privacy notices, we will all accept the contractual conditions in question without reading anything at all.

If we want to avert the risk of a massacre of rights, therefore, it is, first and foremost, essential to identify, at least, new and more effective forms of acceptance of contracts in the digital dimension that are capable of ensuring actual awareness on the part of those who decide to adhere to a contract for access to the metaverse.

Certainly, it will not be enough if, as is likely, being in the metaverse will be for most of us an almost indispensable necessity, much like the use of many digital services today, and if there will be only one metaverse or a small number of metaverses in which it is "worth being."

In substance, in fact, we risk not having a real freedom of choice to exercise.

But in the search for some certainties around which to build a first reflection on privacy, the metaverse, and the future, it seems important to also pause to reflect on another issue: the metaverse - whatever it is or will be - will represent - and, in fact, already represents - an experience enormously more immersive compared to today's Internet in which we are already living somewhat immersed.

This immersion, in the case of the metaverse, will essentially be due to the fact that the metaverse will be three-dimensional and multisensory.

These are some characteristic attributes of the dimension we are dealing with that will make it extremely valuable in a multitude of professional fields and, more generally, in human life: I think of medicine, training and education, specialized remote maintenance interventions, research, security, and dozens of other possible forms of employment.

But three-dimensionality and multisensory experience have a premise that is difficult to doubt: to live any kind of experience in the metaverse, it will be essential to share a significantly greater amount of personal data than we currently share to exist in the digital dimension.

And, therefore, this means that the regulation on the protection of personal data, in the dimension of the forthcoming metaverse, will be even more central and relevant than it is today.

Even today, in fact, in the digital dimension, we are our data because it is our data that represent us, shape us, make us known, tell our stories, and are the basis of every relationship we have in personal, professional, commercial, and political dimensions.

And all this will be even more true in the coming years in the forthcoming metaverse.

What to do in the face of these certainties in the attempt and with the hope of governing the phenomenon that awaits us? It is difficult to answer this question while the contours of that phenomenon are still evanescent, blurred, not very clear, and constantly evolving.

But some hypotheses can probably already be ventured.

The first is that it is essential for public authorities to progressively reclaim at least part of the roles, competencies, and functions that they have ceded to private entities in the governance of the Internet because if this does not happen, we will inevitably find ourselves all guests of a more or less long series of private gardens in which the rules of personal, professional, economic, and political relationships will be dictated by the markets in the name of profit.

It is a bleak scenario for rights and for humanity.

The second is that it is necessary to rethink entire regulatory systems - including that concerning the protection of personal data - within which it has been thought so far to rebalance the positions of power between weak and strong entities by imposing on the latter a more or less stringent series of information obligations.

These obligations no longer work or, at least, do not work as they are currently formulated, in a more formal than substantive dimension: no one reads the terms of use of digital services and platforms or the privacy notices anymore.

What is happening, therefore, is that guarantees introduced into our legal system to rebalance positions of power are producing the opposite effect, making the strong stronger and the weak weaker.

The former can defend themselves against the complaints of the latter by claiming to have provided all the necessary information according to the law, and the latter cannot contest anything further having, in some way,

chosen not to read but to proceed with the selection
their proposal that it was about
adhering to a contract or providing this or that
consent to the processing
of their personal data.

A tool for equalization has, in short, become
a tool
for inequality.

And if the course is not reversed, the fate is sealed:
in the metaverse, even
more than what already happens, we will be free to say
and do what we are
allowed to say and do not only based on rules
written in a law firm and
not in Parliament but on rules that we do not even
know.

And there is, among those that can already be
glimpsed on the horizon,
one last issue of no small importance: being
in the metaverse will mean

exchanging data for services, sharing pieces of the mosaic of our personal identity with other users and with the managers of platforms and services, it will mean, in short, definitively accepting the idea - which we have not yet accepted - that data, including personal data, are legal-economic goods equal to any other, equal to money, gold, silver, or oil.

But we know - or at least, we should know - that it is not so because personal data are also and, indeed, above all, pieces of our personal identity and, in this sense, fragments of a fundamental right that is

a guarantee of other rights and freedoms.

What to do, then?

Too early to say without the risk of being wrong but, probably, if it can be discussed that certain common personal data may - or perhaps even should - circulate freely even in market dynamics, this cannot be true for sensitive data, for those related to our health, our sexual habits, our religious beliefs, our political orientation, for biometric data, genetic data, or judicial data.

These data must be resolutely removed from the market.

And, likewise, it must be prevented that the most vulnerable, starting with children, unknowingly dispose of their personal data in the market dimension.

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None of this, of course, will be enough to make the immersion of the world into the metaverse an experience free from those risks that represent the other side of any innovation, but these are precautions that could, at least, help reduce the risks that serve as a necessary counterbalance to opportunities.

But there is one aspect that is more important than any other:

mass education

about the metaverse.

It is an indispensable fact, an essential precondition for collective benefits to be greater than harms, and it is something that has been lacking in the face of the Internet revolution that has given - and is giving - the world so much but in a heterogeneous manner: much more to a few - those most educated in digital - and much less to many, those less educated in digital.

And, above all, it is leaving many completely behind, making them second-class citizens, relegating them to the role of supporting actors in a society in which they should, of course, be protagonists like everyone else.

We cannot allow this to happen again, to happen even in the metaverse.

And in this sense, there is a need for an authentic declaration of war against digital illiteracy, comparable in size, determination, conviction, and objectives to that which, even in Italy, was declared in the post-war period against literary illiteracy: special laws, adequate economic resources, tens of thousands of courses made available to ordinary people, the media, both old and new, involved to sow culture.

If we do not accept this idea, the society that awaits us and its opportunities, once again, will not be for everyone but only for some, and the future will be less democratic than the past that we will leave behind.

THE PROTECTION OF DATA. FOR 25 YEARS THE COMPASS OF THE FUTURE

Fourth Session

the Era of
cyberwar

- SPEECH BY Sen. Adolfo Urso
- CYBER-RESILIENCE: THE AGENCY FOR NATIONAL CYBERSECURITY
- DEFENSE AND CYBERSPACE

Dr. Annunziata Ciardi

Gen. C.A. Carmine Masiello

Col. Marco Menegazzo

- GLOBAL CHALLENGES OF CYBERCRIME, HOW TO DEFEND ONESELF

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FOURTH SESSION*
THE ERA OF CYBERWAR

Sen. Adolfo Urso
President of COPASIR

I thank the friends of the Authority who invited me to this meeting, an Authority that has been heard by our Parliamentary Committee for the Security of the Republic precisely regarding issues related to cybersecurity.

the current minister Guerini, because

If we consider the start date of my term as the beginning of the legislature, when I entered Copasir as vice president, then the president was

the Democratic Party was in opposition, we can say that at that moment we realized that

Italy was very far behind compared to other countries, and precisely for this reason, Guerini decided and the Committee agreed to conduct the first exploratory investigation of the legislature, which concluded with a Report to Parliament on cybersecurity and information protection, which was the first Report of the Committee to Parliament in this legislature.

In that Report, we identified the issue, the delay in our Country and we explicitly requested, for example, the extension of the Golden Power to the 5G telecommunications sector, which occurred in a short time, just as we requested the prohibition of Chinese technology in the core of 5G, which has still not happened. In the same Report, we also requested that the delay compared to other European countries, especially France and Germany, be addressed with the creation of an Agency or an Institute; the formula of the Agency for national cybersecurity was then chosen, which unfortunately occurred with some delay.

Therefore, when I was appointed to the Copasir, the situation was particularly concerning. Over the years, thanks to the impetus from the Committee and that initial Report, many gaps have been filled, and Italy has increasingly equipped itself better. If we take the date of my appointment as the starting point, which was on June 9, 2021, there is also a significant coincidence. I remember that the first act that came to me, while we were still in the polling station, was the decree for the national cybersecurity agency. The pre-Council had approved it, and the next day it was up to the Council of Ministers; we realized at that moment that there was no oversight Authority; the text approved in the pre-Council did not provide for a parliamentary oversight Authority. So we confronted the Government, we convened the Undersecretary for Intelligence Gabrielli, who came to the Committee before the Council of Ministers, and we discussed what the best approach would be.

The text then approved by the Council of Ministers, unlike the one approved in the pre-Council, identified the Copasir as the oversight Committee for the newly established Agency, so that it is now under our control, just like the Intelligence Agencies. This is because the Copasir, established by a parliamentary initiative law approved unanimously in 2007, unlike the previous body which was COPACO, does not only deal with oversight of the operations of the secret services and the Government's operations in the field of Intelligence, but as its name suggests, it oversees the activities of Intelligence, the Government's actions in the Intelligence field, but also the Government's operations in all areas of national Security. For this reason, from the very beginning, Cyber has been considered a fundamental element of national Security, and the Agency for national Security, although not an Intelligence Agency but a secular Agency operating in the field of public Administration, has been subjected to our oversight.

Because indeed, the Committee oversees, in a broad sense, national Security. We have heard several times from Director Baldoni and also from Deputy Director Ciardi regarding competencies that fall within their respective areas, as well as from the director of the Postal Police. The analysis concerns the global context in which Italy operates to protect its national Security, the privacy of its citizens from hacker incursions, which are not only incursions of a war but also incursions of criminal activity, as they can be carried out for extortion purposes, for terrorist purposes, but can also be configured as state aggressions.

The theory that Cyber could be used for state power or state aggression is a theory that has been developed and was first evoked by the current Chief of Staff of the Russian Defense, Gerasimov, when in 2013 he developed a military doctrine, published in specialized journals, essentially configuring a permanent war by all means, to assert Russian power against the West. In this permanent war, Gerasimov cites not by chance the Information War, that is, a war that essentially aims to influence public opinion using cyber means. Exactly in 2013, a non-coincidental date since in 2014 the first Russian invasion occurs leading to the annexation of Crimea and the creation of the so-called independence republics.

In developing this theory, which then also becomes practice, its substantial aim is to take absolute control of the internal Network. In 2013, when Gerasimov elaborated the military doctrine of permanent war or hybrid war, the internet was considered the vehicle of freedom, and even on the Russian internet, most of the news is...

referring to the desire for freedom:

Russian citizens published news that spoke of their aspirations, to assert their desire for freedom; this also happened, though to a lesser extent, in China.

Well, the regimes realized how dangerous the expansion of the right to freedom that the Internet allowed individual citizens using various online tools was for them. For this reason, in Russia as in China, with different systems, total control of their Internet was first established, and today in Russia, almost all news that circulates on the Internet is news that evokes the narrative of the Regime, that is, the press release that is constructed, developed at the Kremlin, and propagated on the internal network, but then thanks to trolls, fake news, and false social profiles, also disseminated externally.

This has happened simultaneously in both Russia

and China.

One only needs to look at what the national security law says regarding the control of information that is conveyed on the internet; it requires someone to certify who can publish and what can be published: total control of internal public opinion, the latter also to suppress any form of dissent. They have built an offensive machine to penetrate Western democracies, influence them, and pollute them, to poison the wells of democracy, which are democratic debate, freedom of information, pluralism, and in some way subjugate Western democracies.

There are also other tools, such as hacker attacks, the blocking of power lines in some American states, perhaps also for extortion purposes, but there can also be a state threat; if the water supply is blocked in Israel, rather than in the United States, it certainly causes greater damage than a missile attack.

Precisely because hacker attacks are sometimes disguised as extortion attacks by "common" crime; but they are state attacks. For this reason, the Atlantic Alliance has included cyber as a war domain.

When we study history, we realize that some millennia ago, the war domains were two: land and sea.

Over time, and perhaps it was an invention in some way Italian, the sky was included as one of the war domains. After these three domains, space was included, and then cyber was added. The latter is the most pervasive because through cyber, the other domains that absolutely need cyber are invaded.

This means that a large-scale attack on one of the Atlantic Alliance countries can be configured as an attack of another nature on the basis of which to activate the so-called "fifth article."

For this reason, I believe that the Atlantic Alliance and we as a country must equip ourselves in an even more significant way not only to prevent with Intelligence, protect ourselves with the Agencies and other forms of defense that we have, but also to understand how, in the face of a cyber attack that

could be configured as a state attack, we can react, according to the regulations that we must share with the other countries of the Atlantic Alliance. This is the issue that we all must somehow adapt to, because unlike other domains, the cyber domain concerns each of us; the enemy can enter through my cell phone, through the PC. Each citizen must assume that awareness and equip their daily, home defense in order to protect the common space of us all. We all hope that the war in Ukraine ends and that Putin stops the destruction that is ongoing, but we must know that even if this happens, the permanent war that began in 2013 will not end the next day; it will continue in a different form or perhaps increase in intensity in other domains. Let us also prepare for this.

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CYBER-RESILIENCE: THE NATIONAL CYBERSECURITY AGENCY
Dr. Annunziata Ciardi
Deputy Director General of the National Cybersecurity Agency

We pay a price for a revolution that perhaps has no equal in history. This has been a revolution not only technological, of course, and not only cultural; in some way it has been an anthropological revolution that has placed us in a new relationship with the existing, with the real. The digital dimension "involves us all and in everything." This is the fundamental characteristic of this cyber space that has entered every interstice of our lives. We now constantly move on a digital thread even when we are not perfectly aware of it; indeed, even when we believe we are not in any way exposed. The digital dimension is a dimension of which today no country that wants to compete on the international scene economically, socially, culturally can do without.

****Digital Transition is a Fundamental Necessity****

for a country that wants to run, but all this also has a downside: the security profile. It is a very important profile and in the digital dimension, it is an extremely delicate profile. The digital society is decidedly high-performing but very vulnerable: it has many aspects of delicacy and, moreover, in this last period, these have been absolutely accentuated by the two black swans that have struck the globe: first the pandemic and then the war. When the drama of the pandemic occurred, I was in another professional life and was fighting cybercrime with the Postal Police. I had the opportunity to see firsthand the exponential growth of all cyber crimes, while conversely, traditional crimes underwent a significant contraction. Robberies had almost been reduced to zero, especially during the hard lockdown, as well as muggings, etc., while cyber crimes skyrocketed with chilling percentages.

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Then the war, which has triggered a whole series of problems, because today war is also fought in the cyber domain. To all this, we must also add that the situation in our country was not among the most rosy. The National Cybersecurity Agency has been accelerating its steps significantly since June, but it is evident that a structural and complex problem like the issue of digitalization and the consequent security problem cannot be solved in the span of a couple of months. Countries much better equipped than ours, in terms of digital security, have recently faced numerous problems. Let's think of the United States with the Pipeline incident or Israel hit by DDoS attacks against its main sites, and we are talking about two countries for which digital security has been cultivated in an absolutely preferential manner and certainly much earlier than in our country. The Agency was established in June 2021 and became operational with the first two employees in September. We are talking about just a few months. Currently, we are about a hundred people and we should, by regulation, become 300 by next year and 800 within four years.

Therefore, an accelerated progression is planned precisely to ensure the resilience of the country. Obviously, the agency does not arise in a desert of initiatives; however, a fundamental piece was missing: that of resilience. Cyber resilience must ensure that a country, even in the event of an attack, can not only work to prevent it, to increase the defenses of critical infrastructures, i.e., those that provide essential services for the country, but also be able to react quickly once it has been hit. The agency was created to counter all these threats that are more widespread, more diversified, and more sophisticated through a whole series of elements. In Italy, according to data we have from law enforcement, in the last year significant cyber attacks, therefore not all cyber attacks, but only those directed at critical infrastructures, i.e., towards the essential services of our country, have increased by 35%. We are talking about enormous percentage increases because technology evolves rapidly.

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We must somehow absolutely balance the technological transition and the digital transition in a sustainable growth of the digital country. We need to equip ourselves with capabilities, institutions, resources, and skills useful for increasing the resilience of the country. I was asked if the number of Cyber soldiers we have in the country is adequate. I respond without mincing words: "No!" In our country, a bit all over the globe in fact, the professionals are lacking. One of the tasks of the Agency is to stimulate the growth of professionals useful to counter the cyber threat, the threat directed at digital companies. It is a type of preparation that has not been sufficiently cultivated. Those few extremely valuable resources we had often chose in the past the path abroad. Also for this reason, one of the tasks of the Agency is to stimulate the growth of these professionals, to encourage relationships with universities so that they implement a virtuous circuit so that the preparation of our country can draw on useful professionals in this field. On the other hand, the Agency has a whole series of tasks that could not be implemented without professional figures. On one side, the prevention and mitigation of attacks that have already occurred, on the other, technological independence.

When we talk about security, we cannot remain silent about a problem of total technological dependence that not only our country but I would say the entire European continent suffers from. We

are dependent, despite being large users of technology, on countries that are much more advanced than us in this field. Technological independence is not only an economic development driver that is extremely important at this moment but also a security factor. It is undeniable that having our own technology protects us, allows us to manage certain aspects better. The Agency must somehow stimulate the growth of European technological independence first, in collaboration with other European countries, and ultimately also that of Italy. Moreover, the growth of awareness in the country is important. The “care for security” must be domestic, of the individual, an awareness that must involve each of us. A company, a region can also invest a lot.

the awareness must be for safety, but if an employee then takes their company workstation home and violates fundamental cybersecurity rules, it is evident that they put all those measures that the company has painstakingly tried to establish in serious danger. Therefore, investing in each of us is crucial, and understandably, we all struggle to have this awareness precisely because the digital revolution has been rapid. Just think that in 2009, only 15% of Italians had a smartphone; the phone was used only for calling. Today, in just a handful of years, over 129% of Italians have a mobile device available, potentially connected 24 hours a day. This illustrates how much technology and digitalization have experienced tumultuous growth, and it is obvious that it is challenging to metabolize such a change, to grasp the importance of adopting security models that are now fundamental in our lives and in the protection of our data that is online and constitutes our life online. Digital security, cybersecurity, today is increasingly the comprehensive security of our lives.

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DEFENSE AND CYBERSPACE

Gen. C.A. Carmine Masiello

Deputy Chief of Defense Staff

Question from the moderator Dr. Fausto Carioti: So let's talk about soldiers, and it happens that next to me is General Masiello, Deputy Chief of Defense Staff. General, good evening. I also ask you to explain to us what your institution does, what role it plays within cyber warfare? And then I ask you what has changed since February 24?

Gen. C.A. Masiello: Thank you, I greet everyone, I greet the Authority. First of all, thank you for inviting the Defense to this meeting; it is a sign of attention not only for the world it represents but also a sign that it has been understood that the Defense has an active role in this cyber world, and it is one of the actors, let me say, fundamental. This relates directly to the first question you asked me when you inquired about what I had found in my work: I have been dealing with cyber in a broad sense for about 5-6 years, since I was assigned to Palazzo Chigi as military advisor to the Prime Minister, and at that time I chaired the cybersecurity security nucleus, so that was my first contact with the cyber world. And I must say, echoing a bit the words of President Urso, that even back then it was evident that the situation in our country was not optimal and that measures needed to be taken, things that have since been done over time with the creation of the CN and the perimeter, etc. Now I am responsible at the Defense General Staff, through the units under my command, for defining the cyber policy of defense and developing new capabilities in the defense sector. However, the defense is not new to the cyber phenomenon; for us, the issue of network security, let's say, is somewhat endemic and is part of our training. We are a bit obsessed; we grow up with this security issue, and therefore we are somewhat advantaged compared to the rest of the Community. Secondly, we are used to talking with our allies; we are accustomed to looking at what happens in other countries, in a manner that I would say is almost exaggerated, and this is not only within the framework of alliances but we look a bit all over the world. The third point, let me say, is that we have studied the Gerasimov doctrine: for us, it was important to understand what was happening on the other side of the barricade, if I may use that term. Therefore, in light of this situation, we have restructured our entire governance, and this is in line with what was happening in the rest of the country; thus, we essentially have two levers, one lever that falls under my competence and then an operational lever that is to actually conduct cyber operations, so this is a bit of a summary of what the defense does. I would like to highlight a few points. The first point

has already been addressed, and I will say it immediately: it is sharing, synergy, as I call it. The cyber issue cannot be addressed separately by the individual actors in this country. It is necessary to make an effort that is difficult in Italy, an effort to look at the problem in a coordinated manner; therefore, synergy in activity is required, and above all, sharing is needed because in the cyber world there is little willingness to share. There is little willingness to share because there is a reluctance to show that one has been breached or to show that one has made a mistake, and there is a reluctance to show that perhaps one has something more than others. And this is part of the rules of competition. But this is the worst thing one can do, because the fact of not sharing with others the problems that have been suffered inevitably leads or can lead others to suffer from the same problem, and therefore one must make an effort in this direction, which is not always easy because it is a cultural issue.

Regarding February 24, because this perhaps is the point that most pertains to the work we do: the operational work has certainly marked a watershed, the first thing I feel I must say is that February 24 marked the beginning of the first symmetric conflict of the digital age, in the sense that we have lived the last twenty years, and I speak of officers of my generation and those more recent, we have lived for twenty years in combat in Iraq, Afghanistan, to speak of the main theaters, but it has always been about asymmetric conflicts, that is, between units that had totally different technological capabilities.

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In fact, I have been in Afghanistan and I must say that the cyber issue was really relative. It was not felt compared to what happened in Ukraine. This is the first factor that should make us reflect: from this moment the cyber aspect has entered into war in a broad sense because we are talking about war and because it is a cybernetic conflict.

Carioti: Excuse me, what do you mean by symmetric?

Gen. C.A. Masiello: Symmetric means that they have the same capabilities. They are two adversaries that operate at the same level of capability, we and the Russians would be a symmetric conflict. We and the Taliban: it is an asymmetric conflict, so to simplify. And so I was saying: in Afghanistan, this was not discussed. And I was saying therefore: cyber conflict, because it is a real conflict, we must give words their meaning of a daily conflict. Every day in the world there are millions of cyber attacks and it is no secret, most of them kept below the threshold, obviously to avoid a response from the attacked party. Cyber then, as has already been said by President Urso, is now part of the domains in every respect, so we have seen in the conflict of February 24 how the cyber domain has been integrated into military operations, so cyber attacks were conducted in parallel with conventional attacks.

So while tanks were moving and planes were bombing, there were cyber attacks being conducted against Ukraine in particular, and not only. And we are clearly talking about hundreds of attacks. And this has certainly been a great lesson for us, so we knew how it could be. We knew in theory what could happen, then we saw it practically. Another very interesting thing is the issue of space, because space as a new domain, has been mentioned, is closely linked to cyber. Not ensuring cybersecurity in the space domain evidently renders space useless, we rarely think about this, we do not pay much attention because we are used to it, but if we think about living 10 minutes, half an hour, an hour without space, we probably would not be able to live anymore.

Today the positioning we have, our cell phones when we move, is all provided by GPS, by satellites - positioning that is in orbit. Communications from one continent to another occur thanks to satellites. Financial transactions occur thanks to satellites. So the satellite images, which have played and are playing a huge role in the conflict in Ukraine, that too would stop if there were no cybersecurity. So these are two domains that move in parallel. A lot of attention must be paid to this; but it is something we do not do, we did not do it with the Internet, we did not do it with the growth of digital, we have not been able to regulate it, we have not reflected on it, we have not made a cultural shift, the same thing is happening in space where we are approaching with the same lightness. Just as we thought that the internet at first was merely a field to exploit, perhaps a new domain for the economy.

In fact, we talk about space economy, it is the most well-known term when talking about space, but no one cares about security. I always give this example when I speak about these domains, this example is: Space and cyber are a bit like the Mediterranean Sea, that is, the dimension of the Mediterranean Sea, where progress depends on security, that is, if the Mediterranean is safe, if ships can move, if trade can occur with the different shores safely, without submarines attacking us, this guarantees our progress.

Here, space, just like the cyber domain, is a bit like that, we must ensure its security so that progress can evolve, because we are talking about progress, we are talking about the growth of our society. The future (we are already in the future, this has been said) we as defense share the same problems that the cyber agency shares, obviously.

The main problem is a lack of resources. Israel has been mentioned, Israel is a beautiful example, unfortunately not fitting. Why Israel? As everyone knows, it has a conscript army, so all Israeli citizens go through...

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through the leverage, the enormous basin is clear, but above all, you must necessarily pass through there; then if you are good, they will take you. We obviously do not have this opportunity, and therefore we must try to attract the best young people to come to our Institutions, and this is not difficult, it is not particularly difficult. The problem that I personally consider essential is that many times everything is reduced to a salary issue. How much do I earn here, how much do I earn in the private sector, how much do I earn in the public sector. And perhaps instead we should leverage values, what it means to work for your country, to work for your flag, this is very important. And it is not usually an Italian thing, because in all other countries, when working in the public sector, you earn less than in the private sector.

However, other countries know how to leverage more on the spirit of the homeland, on the love for their country, and this helps them recruit these people. So for us, it is a significant effort in recruitment and a significant effort in training, training not only for the newcomers or those who are already there, but especially training for our young leadership, those who are approaching a military career because they must be accustomed from a young age to think cyber and to think space; I always put them together, because only through the integration of all these capabilities can they become the commanders of tomorrow and can guarantee the defense and progress of our country. Thank you.

DATA PROTECTION. FOR 25 YEARS THE COMPASS OF THE FUTURE

THE GLOBAL CHALLENGES OF CYBERCRIME, HOW TO DEFEND ONESELF AND THE ROLE OF THE GUARDIA DI FINANZA

Col. Marco Menegazzo

Commander of the Privacy Group of the Special Unit for Privacy Protection and Technological Fraud of the Guardia di Finanza

1. INTRODUCTION

First of all, I would like to extend my personal thanks for this opportunity offered to the Guardia di Finanza to present its testimony and point of view on highly topical and particularly operational issues, such as those related to the world of cybersecurity and, more generally, to the world of national security in the face of threats posed by modern technologies. I am Colonel Marco Menegazzo, Commander of the Privacy Group of the Special Unit for Privacy Protection and Technological Fraud of the Guardia di Finanza, a department based in Rome with national competence, primarily oriented towards collaboration with the Data Protection Authority, for operational profiles concerning the protection and safeguarding of personal data and combating cybercrime, regarding aspects related to cybercrime that have repercussions on the economic and financial interests of the Country.

During my speech, I will outline the main informational elements concerning the context of cybercrime and the consequent need for defense against attacks on data and information systems, the relationship between cybersecurity and privacy protection, and the role of the Special Unit for Privacy Protection and Technological Fraud as a countermeasure deployed by the Guardia di Finanza.

2. CONTEXT ANALYSIS

In the current context, characterized by the exponential increase in the use of IT tools and the parallel increase in attacks on networks and information systems, the cyber domain represents a "battleground" with implications for the regularity of the provision of essential services and national security, and, not least, for the economic and financial interests of the Country, whose protection represents the institutional mission of the Guardia di Finanza. Legislative interventions, through the NIS Directive, the establishment of the national cybersecurity perimeter, and the creation of the National Cybersecurity Agency, in addition to harmonizing with European regulations in this area and with similar initiatives undertaken by other EU countries, aim at the progressive growth of the culture of cybersecurity, whose crucial factors remain the strengthening of collaboration between the private sector, central administrations, and local institutions, within a framework of greater promotion and knowledge of digital culture also for the industrial and technological development of the Country. The national strategy, as indicated particularly by the Directive, provides for the definition of plans for assessing cyber risks, raising awareness, and training citizens in cybersecurity, and above all, it provides for the preparation of countermeasures to cyber incidents in terms of preparation, response, and recovery of potentially compromised services, within a framework of pronounced resilience of the strategic information systems for the Country.

In accordance with the provisions of the Directive, it is evident that the information systems that operate essential services for civil life must necessarily be equipped with "adequate" technical-organizational measures for risk management and for the prevention of cyber incidents. However, they must be capable of reducing the impact of any incidents that occur by demonstrating adequate resilience through the timely restoration of any compromised services. In fact, if a service is fundamental for the maintenance of social and/or economic activities, if the provision of such a service depends on the network and information systems, and if a cyber incident produces significant negative effects on the provision of such a service, the repercussions in terms of functionality, frustration of the rights and legitimate expectations of citizens, economic damage, and reputational harm can be considerable.

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The "cyber world," which we might define as that set of interconnected technological infrastructures, including the internet, that extends to all technological devices (telecommunication systems, computers and their components, hi-tech products, etc.) capable of connecting with each other, represents a new dimension in which technologies increasingly characterize and condition, in a decisive manner, the lives of modern societies, expanding the perimeter of what until now we have called the real world. The cyber world consists of physical structures, mathematical processes, software, and information, which ultimately represent an articulated and complex system, not always fully known in all its parts, considering the networks of the so-called "dark web," capable of generating significant profits but also substantial losses for national economies and systems. The aspects related to the global network increasingly touch the daily lives of people, reverberating countless implications in the workplace and social spheres, but in many cases, there is still a lack of awareness and understanding regarding the opportunities but especially the risks. The relationship between citizens and technologies sometimes highlights concerning aspects regarding literacy and the correct use of electronic devices, often revealing criticalities and behavioral vulnerabilities that provide fertile ground and profit for numerous illicit activities. Similarly, a comparable situation is found on the side of businesses, where a poor perception of the problem and insufficient technological sensitivity, often accompanied by the absence of dedicated skills or organizational roles, end up paving the way for cyber threats or fraudulent actions, which are increasingly massive and sophisticated. As indicated in

the Censis-DeepCyber report on cybersecurity in Italy, presented last April 22 in the Senate of the Republic, the situation is mixed. In fact, while 61.6% of Italians are concerned about cybersecurity and take precautions on their devices to protect themselves, the remaining 38.4%, almost 4 out of 10 Italians, are indifferent or do not protect themselves from cyber attacks.

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On the other hand, 81.7% of Italians fear being victims of theft and violations of their personal data online, perceiving the risk mainly while browsing the web, using social accounts, purchasing products online, and conducting home banking operations, but only 1 in 4 has a clear idea of what cybersecurity is. The global cyberspace, of virtually infinite dimensions, is extremely complex, especially when viewed from the perspective of systemic risk. It also constitutes a place of interest for organized criminal networks, whose objective is to steal money, defraud or deceive citizens and organizations for profit, thereby distorting fair competition, a new and unprecedented battlefield and area of geopolitical competition. All this results in enormous repercussions on the national economic and productive fabric, to the extent that the vulnerability of systems and networks, even from the perspective of economic/financial security, is among the fundamental factors of concern for the majority of institutions and for economic and social categories. Particularly insidious are the new criticalities due to the exponential growth of all those illicit activities that find particular ease in the deeper part of the network, in that hidden area of the dark web, which is not indexed by search engines and offers ample guarantees of anonymity thanks to the numerous hidden services present there and the massive use of financial transactions conducted through cryptocurrencies with intrinsically anonymous characteristics. The emergency situation created by the Covid-19 pandemic has led to a massive recourse to digital means, allowing many companies to continue their activities, as well as enabling the Public Administration to continue providing services to citizens. With smart working, the new mode of work imposed by the Covid-19 emergency, the attack surface for cybercrime has expanded, opening new opportunities for cybercriminals: employees have often proven vulnerable to phishing campaigns aimed at capturing access credentials to information systems, thereby exposing corporate data or sensitive data to the risk of loss or theft.

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****Transformation, Where****

It is therefore a particularly complex and continuously evolving framework. The increase in opportunities is accompanied by a parallel increase in vulnerabilities, requiring ever greater awareness and attention from all concerned institutions. The massive spread of technology makes life easier but at the same time also more fragile and vulnerable: the issues related to digitalization and the dissemination of a culture of cybersecurity and resilience did not arise with the pandemic, but were and remain central. The challenge concerns cybersecurity, which can no longer be considered a cost or a field for experts only; it is increasingly becoming a social investment of collective interest, essential for a proper digital evolution.

****In Security of Infrastructures****

The liberation from technological dependence, the awareness and training of users within the framework defined by the roles assigned to the National Agency, the investigations carried out by the Police Forces, cyber intelligence, and cyber defense.

Nonetheless, the need to intervene in cybersecurity, with appropriate investments and within a coherent framework of needs and priorities suitable for addressing global cyber threats, makes it unavoidable to implement the National Recovery and Resilience Plan (PNRR), which provides for specific projects in the field of cybersecurity, in addition to specific initiatives for digitalization and innovation as tools to relaunch the competitiveness and productivity of the country system. The role of digitalization and innovation indeed concerns, transversally, all the Missions provided for by the Plan: Digital Transformation is therefore a mandatory step to enable Public Administration, as well as private

companies, to face the challenges posed by the current economy.

****Parallel to the Challenges Posed by Cybercrime****

In terms of aggression against public and private information systems, as well as to the detriment of citizens, the issue of the use of virtual currencies in illegal cyber activities arises, whose connection with the institutional mission of the Guardia di Finanza emerges clearly both from the perspective of judicial police for the prevention and repression of crimes and from the economic-financial police for aspects related to the laundering of illicit proceeds and the introduction into the economy of illegally sourced capital.

Virtual currencies or cryptocurrencies, whose origin dates back to 2009 with the creation of Bitcoin, can be used in a wide variety of ways, some legitimate and ethically correct, others illegal and dangerous, posing ever-new challenges for law enforcement operators, which can only be addressed by having clear and defined criteria underlying their essence and functioning. One of the characteristics that makes control and oversight of the cryptocurrency world difficult is decentralization, or the absence of a central authority that can exercise any type of power over them and the activities carried out by users, as cryptocurrencies, by definition, lack the requirement of materiality.

Another aspect to consider is that cryptocurrencies operate through the Internet, thus being devoid of any nationality or territorial reference. These characteristics, in general, can also be found in the other hundreds of virtual currencies that exist, created after Bitcoin, especially in those characterized by a high degree of anonymization.

From a regulatory perspective, there are critical issues regarding the seizure of virtual currency and its conversion into current currency for allocation to the Single Justice Fund. Other issues to consider are those related to the availability of cryptocurrency funds for the execution of undercover operations and the desirable regulatory amendment concerning the extension to cyber crimes of the institution of "Undercover Operations" as per Article 9 of Law 146/2006.

****3. CYBERCRIME AND DEFENSE AGAINST DATA AGGRESSION****

The national regulatory framework for the protection of data security and the understanding of the dynamics that agitate the world of cybercrime cannot...

naturally, on their own, to contain threats and prevent damage arising from cyber attacks, especially those carried out by particularly skilled and determined individuals and organizations, and therefore insidious. Nor can the repressive function exercised by institutional actors, primarily by the Police Forces, avert the emergence of new malicious actors and new technologies for attacking public, corporate, and private cybersecurity. The main barrier remains the prevention and protection of IT infrastructures and data through operational national security policies adopted by Governments, Institutions, businesses, and also by individual citizens, both by implementing active defense measures and by increasingly disseminating the culture of cybersecurity awareness at all levels.

Ultimately, whether it concerns a Public Administration, a private enterprise, or ordinary citizens, the very definition of cybersecurity provided by the Law and in particular by Article 1 of Decree 82/2021, which established the National Cybersecurity Agency, states that the set of activities necessary to protect networks, information systems, IT services, and electronic communications from cyber threats must ensure the availability, confidentiality, and integrity of data, guaranteeing its resilience.

Ultimately, it is the concrete response to these guiding principles that constitutes the only type of effective defense against attacks on data. Essentially, information must be accessible at the specific moment it is needed, exclusively to those entitled to access it, and must not be altered or corrupted. These characteristics must also be guaranteed throughout the entire lifecycle of the information, that is, from when it is generated and/or collected to when it is destroyed, obviously passing through every type of storage. In addition, resilience must be considered, meaning the ability of systems to withstand an attack not only in terms of the "robustness" of the structures but especially as the capacity to recover the intended functionalities as soon as possible through the timely restoration of full operational capability. It should be emphasized that these programmatic reference points can be compromised not only by hostile actions via malware, a typical battlefield of cybercrime, but also by the

possibility of physical attacks on information systems in a symbolic, propagandistic, and political context. I am referring to potential terrorist acts aimed at the infrastructures where processing systems reside and the consequent unavailability of data and services of significant or strategic importance for the Country. Defense against these threats is possible with adequate security measures and physical surveillance of infrastructures but above all by adopting disaster recovery solutions capable of preventing data loss and restoring functionalities quickly and in any case without impacting the productivity of the systems potentially involved. The same optimization of disaster recovery solutions in terms of operational capability applies to all physical threats, namely those natural or accidental events, such as earthquakes, floods, fires, and reckless human actions, that can compromise the physical integrity of systems and data. More complex but no less critical from the perspective of responding to aggression against public or private informational assets is the world of cybercrime as traditionally understood. As is known, cybercrime is the criminal phenomenon characterized by the abuse of information technology. Cybercrime can generally be defined as a criminal activity that involves the structure of information technology, including illegal access, interception, interference with data, misuse of devices, identity theft, and electronic fraud. Given the transnational dimension of cybercrimes that operate in the extraterritoriality of the Internet, the need for international collaboration has materialized with the approval, by the Council of Europe, on November 23, 2001, of the so-called Budapest Convention. This represents the first international agreement on crimes committed via the Internet or electronic networks. With this Convention, the adhering States have adopted legislative measures aimed at the criminal repression of cybercrimes, thereby harmonizing the various domestic legal systems and coordinating forms of collaboration for the purpose of evidence acquisition. The ratification of the convention has modified the criminal code, with...

the introduction of new specific types of crime and the modification of certain already provided types, the code of criminal procedure, with reference to fundamental articles for police investigations, the competence of the District Attorney on cyber crimes, and the administrative liability of the company under Legislative Decree 231/2001 for certain cyber crimes.

Judicial police investigations therefore represent a response to data breaches, at least in a repressive sense aimed at targeting the perpetrators of criminally relevant harmful events and in a preventive sense of deterrence and potential discovery of new threats to be faced with appropriate countermeasures. However, cataloging possible technical attacks is a difficult exercise, given that their quantity and quality are characterized by the creativity and ingenuity of attackers, and often, in real attacks, different techniques can be used in combination, making a possible classification more nuanced. Generally, the term "malware" refers to the particular category of software programs whose purpose is to harm the user or the systems on which such programs are inadvertently loaded, by executing unexpected and unauthorized processes. There are different types of malware, but their composition and continuous evolution make a coherent classification difficult.

The task of malware is to infect target computer systems in order to damage them or to extract various types of information, especially confidential or sensitive information, particularly from an economic standpoint. Other forms of cyber-attack involve exploiting vulnerabilities in networks, base software, hardware and software configurations of systems, or flaws in IT organization to gain uncontrolled access to the systems themselves. In many cases, such access is facilitated by the users themselves, who are induced to reveal the necessary information, such as accounts, passwords, IBAN, through techniques that essentially rely on the vulnerability of the "human factor." In this case, cyber attacks exploit any form of human weakness or vulnerability of the victims, leveraging increasingly sophisticated techniques of "social engineering."

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Therefore, the response to the question "how to defend against cyber attacks" also and perhaps especially involves a series of precautions and good general practices that should be the heritage of all citizens, both in their capacity as members of more or less complex public or corporate organizations, and in the private sphere. Virtuous behaviors are ultimately fundamental but are no less effective if

practiced consistently and with vigilant attention. The following rules are referred to:

- perform periodic backups with the 3-2-1 rule, where 3 stands for three versions of the data, one original and two copies, 2 stands for two different types of storage for the copies, and 1 stands for one copy kept in a different location;
- use complex and possibly multifactorial passwords;
- use perimeter protection firewall systems;
- use institutional emails with enabled encryption;
- avoid compulsive clicking and verify the senders of email messages;
- protect workstations and mobile devices with antivirus;
- update base software when required;
- adjust protection plans and develop new systems with the philosophy of "security by design";
- periodically update on security solutions, including consulting thematic channels.

4. CYBERSECURITY AND PRIVACY

Thus, there are various threats to which the security of the cyber dimension is constantly exposed, in which conflicts are increasingly shifting to a terrain essentially devoid of an adequate framework of international law. In this context, the balance between data protection and the safeguarding of cybersecurity is complex but also heralds new challenges and synergies. In summary, it is necessary to manage to balance the needs for cybersecurity protection, as a relevant dimension of national security, with the needs for protecting citizens' privacy, through respect for the principle of proportionality, the sole point of reference for balancing general prevention needs and individual rights. However, it can also be observed that the relationship between data protection and cybersecurity presents points of contact and mutual functionalities. In fact, while cybersecurity primarily involves the protection of data and the infrastructures that make up the digital ecosystem, ultimately extending to national security, any compromises of personal databases belonging to public or private entities, the so-called "data breaches" provided for by the regulations on the protection of personal data, not only assume importance for the individual citizens concerned but can also be relevant for national security depending on the type of data accessed illegally. Especially in the economic-financial field, which is the area of intervention of the Guardia di Finanza, the relationship between data protection and security means simultaneously protecting individuals and the community. To address global threats to cybersecurity, the objective must be pursued from a supranational perspective, at least European. Noteworthy, from a legal, economic, and symbolic point of view, is the applicability of the GDPR to entities outside the Community that process data within the European continent. In this way, the so-called "web giants" have been drawn into the orbit of Community law regardless of the location of their business activities. On the other hand, in a space devoid of a "physical" dimension like the internet, sovereignty must be inspired by new forms, not necessarily linked to the traditional criterion of territoriality, capable of ensuring that citizens can effectively protect their rights. The manipulation of personal data and the profiling of users aimed at biasing the will of citizens make evident the related profiles of national security as a possible alteration of orderly democratic competition. Ultimately, online disinformation, exploiting the characteristics of the cyber domino, can condition or influence the social, economic, and political processes of the country. At the same time, the risks to national security must be considered, arising from the adoption of high-value-added technologies by countries where the political system has a strong impact on technology-producing companies, especially in the telecommunications field. Any commercial agreement cannot disregard the respect for the fundamental conditions for the protection of the data rights of European citizens, a protection that, in this case, also has effects on the field of cybersecurity.

5. THE ROLE OF THE GUARDIA DI FINANZA

In the National Cybersecurity Strategy document, presented by the National Agency at the end of last May, among the technical-operational pillars of the overall national cybersecurity architecture, the function of preventing and combating cybercrime is indicated, which is carried out by the specialized components of the Police Forces, thus also mentioning the Special Unit for Privacy Protection and Technological Fraud of the Guardia di Finanza. In fact, the awareness of the serious threats to the country system arising from the illicit use of new technologies has led the Guardia di Finanza to

strengthen its response to criminal conduct that impacts the economic and financial fabric, which is continuously evolving. In this context, the activities carried out by the Corps are articulated, based on the powers attributed to it, in economic and financial police activities, as well as judicial activities, extended to increasingly diversified areas in technological and territorial terms, specifically covering the internet and more generally technologies, and increasingly surpassing national borders in cooperation with countries adhering to various international organizations provided for by individual treaties or conventions. The Guardia di Finanza, in all its operational expressions, operates transversally within the institutional mission of economic and financial police entrusted to it, through an inherently multidisciplinary approach. The action modules, even in investigations that impact the digital world, are therefore simultaneously oriented:

- to the economic control of the virtual territory, through the

****Monitoring of the telematic network, including hidden networks, to verify the existence of pockets of illegality and to intercept "suspicious" financial flows;****

- to verify the tax position of the subjects under investigation for the potential taxation of illicit proceeds;
- to intervene, transversally, on various profiles, such as, for example, those related to currency, intellectual property, product safety, competition and market, as well as privacy.

In combating economic-financial crimes committed on the internet, or through technological devices, the Corps intervenes through two lines that are in continuous functional contact with each other: strengthened by

- the network of territorial departments, distributed throughout the national territory, with the task of ensuring, in their respective areas, the execution of institutional tasks. Among the departments, the Economic-Financial Police Units, with further specific professional skills, serve as leading investigative units;

- the special departments that support the first and are established for the in-depth study of specific matters, are tasked with directly carrying out, or with support actions for operational units, investigative modules characterized by high qualitative standards.

The latter, precisely due to the increasing complexity of criminal phenomena, assume a fundamental role in the strategy to combat this type of crime, especially in terms of "knowledge support," constantly acquiring and updating the knowledge and technical expertise useful for the action of all departments.

In order to better outline the framework of competencies and attributions of the Police Forces in general, with a decree from the Minister of the Interior dated August 15, 2017, the guiding principles of the actions of the structures dedicated to the virtual and technological world of the State Police, the Carabinieri, and the Guardia di Finanza were established.

The prevention and contrast actions, which involve all Police Forces, must unfold on the "virtual ground" where criminal organizations, which have solid roots in the real world, increasingly find a pervasive projection. Nevertheless, the adoption of criteria for the distribution of specialties was deemed necessary, while maintaining the specific attributions of the individual Police Forces and their respective cross-competencies on criminal activities of greater social alarm, such as, for example, drug trafficking.

The Guardia di Finanza, taking into account the attributions of economic-financial, currency, and administrative police conferred by Article 2 of Legislative Decree No. 68 of March 19, 2001, and the specific sector regulations, through the Special Unit for Privacy Protection and Technological Fraud, has primary competence for the research, prevention, and contrast of crimes perpetrated by exploiting the network in the areas of tax revenues, public spending, competition and market protection, currency regulations, and financial markets.

Moreover, the Special Unit for Privacy Protection and Technological Fraud, in compliance with the principle that has assimilated the virtual environment to the physical territory, constantly conducts online investigations, both through Open Source Intelligence methodologies and through sophisticated techniques for gathering, analyzing, and filtering the information found, in order to isolate potential web resources engaged in illegal activities.

The Special Unit for Privacy Protection and Technological Fraud, which depends on the Command of Special Units, is the department of the Guardia di Finanza that ensures constant economic-financial police presence online. The General Command of the Corps, since 2001, had recognized the need to establish a department that would be daily engaged at the forefront in an operational context of exponential technological development, where the interests of criminal organizations had identified a flourishing environment to pursue their objectives to the detriment of the economic-financial sector. Over time, the awareness of a greater commitment in the field of economic-financial crimes perpetrated online led, during 2004, to the elevation of the then Technological Anti-Crime Group (GAT) to the Special Unit for Telecommunication Fraud, later Technological. In 2012, its organizational structure was simultaneously strengthened, providing for the creation within it of four operational groups that develop police activities.

****Judicial Action in Cybercrime****

Constant monitoring of the network aimed at identifying illegal phenomena, including money laundering and financing of international terrorism, is carried out. The cryptocurrency sector is also under scrutiny, as the high degree of anonymity guaranteed by their use makes them increasingly employed for illicit transactions. In 2018, with the advent of the GDPR and a consequent regulatory reform, the Unit integrated its operational structure with actions to protect privacy, increasing from 4 to 5 investigative groups. The Special Unit for Privacy Protection and Technological Fraud also guarantees assistance to other departments of the Corps in all institutional interest sectors where violations are committed using the Internet or so-called "new technologies." In this context, the Unit serves as the Technological Hub for the entire Guardia di Finanza and is tasked with taking a central role in managing technologies and developing IT systems to assist police investigations, providing specific support to territorial units in the field of "Computer Forensics & Data Analysis," as well as exchanging experiences with external institutional entities to initiate specific collaboration initiatives that improve the professionalism and training of personnel employed in these specific service areas.

All of this obviously requires constant professional updating, also through continuous comparison with similar "law enforcement" structures both nationally and internationally, research centers, the academic world, trade associations, and major economic operators in the field of technologies and communication security. It is also worth mentioning that the Special Unit conducts teaching activities with its representatives within university master's programs organized by various universities, as well as at the training facilities of the Corps and participates in thematic conferences proposed by institutions, entities, associations, and national and international police forces. There are numerous participations in inter-force working groups at the national and international level.

Significant and demanding for the protection of legality and security is the monitoring conducted on the dark side of the web, the so-called dark web, where numerous illicit activities proliferate, particularly insidious when considering serious phenomena such as arms trafficking, drug trafficking, counterfeit documents, personal data, etc. The peculiar characteristic that makes the Dark Web appealing for the perpetration of illegal practices is the high degree of anonymity it offers. It is, in fact, a particular protocol of electronic communication that provides various services, such as web, chat, and email services, which are not indexed by common search engines used for Internet navigation.

****6. CONCLUSION****

The articulated dimension that we call cyberspace cannot be defined in its entirety by technological parameters such as connection speed, the number of interactions between users, and the accessibility of data and information online; rather, it is necessary to consider the countless interconnected and communicating services designed to meet the daily needs of our communities and to carry out their related economic activities: I am fundamentally referring to energy infrastructures, financial markets, potable water supplies, mass transportation, and, not least, the essential functions of the State, including its defense and integrity. The duality between the digital world and the real world tends to blur within the complexity and interdependence of systems, often making it problematic to identify their respective boundaries and characteristics.

On one hand, the "migration" towards digital is made more convenient by the relentless evolution of modern technologies; on the other hand, the security of society and, in perspective, the economic development and well-being of the Country can only be guaranteed by the resilience and security of the networks and systems on which such services are based. The elimination of threats arising from cyberspace cannot be fully achieved by any organization, no matter how technologically equipped and procedurally prepared, whether the threats themselves aim to obtain illicit profits, as in the case of cybercrime, or to generate informational advantages or disseminate divisive narratives.

polarizing in adherence to specific ideologies in the geopolitical competition, as in the case of cyber-espionage. The challenges posed, therefore, require the enhancement of the resilience of digital infrastructures, according to an approach that includes the adoption of prevention and risk mitigation measures in networks and data systems and, above all, the increase of awareness among institutional actors, private enterprises, and citizens through a widespread culture of cybersecurity.

It is in this context that the misalignment of the perception of one's security emerges, as the actions aimed at protecting the physical or asset sphere do not, too often, correspond to the equal protection of the digital dimension. Furthermore, the low costs and high availability of offensive tools combined with the increased level of complexity of attacks, the technical difficulty of identifying their authors, and the vulnerabilities of widely used software and hardware, contribute to the current rise in the trend of hostile actions.

The dissemination and progressive use of new technologies, whose impact presents increasingly disruptive characteristics, risks generating multiple potential repercussions in the economic, social, and political spheres, both in terms of technological dependence and loss of strategic autonomy of the State, as well as from the perspective of human error caused by malicious actors characterized by varying degrees of sophistication and driven by different, yet equally harmful, intents.

It is on these premises, therefore, that the "Country System" must continue and increase initiatives in the field of cybersecurity. This is imposed by the new forms of strategic competition that characterize the geopolitical scenario, the protection of national interests, and the citizens. The objective must therefore be the wise construction of a cybersecurity ecosystem, based on collaboration between the public and private sectors, in which the national cybersecurity strategy unites security and development in respect of the values of our Constitution, the commitments undertaken within the international organizations to which Italy participates, particularly as provided for by the strategies of the European Union and NATO.

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Institutions and economic operators, particularly the managers of the infrastructures on which the provision of essential State services depends, the world of academia and research, and civil society must take an active role in protecting their IT assets and become truly resilient to threats, especially in the sectors of ultra-broadband, 5G, and cloud, on which citizens' attention and trust are primarily focused.

I renew my personal thanks to the Authorities present and to the Guarantor Authority for this valuable opportunity for discussion on issues that traverse the generational segments of our society, from a private and work perspective, raising different fears and expectations, but which significantly impact the lives of all.

THE PROTECTION OF DATA. FOR 25 YEARS THE COMPASS OF THE FUTURE

Fifth Session

Information in the Digital Era: True, False, or Virtual?

- MODERATOR: Baldo Meo
- Martina Pennisi
- David Puente
- Raffaele Barberio

- Massimiliano Panarari
- Federico Ferrazza

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FIFTH SESSION

INFORMATION IN THE DIGITAL ERA: TRUE, FALSE, OR VIRTUAL

AGOSTINO GHIGLIA

We now move on to the last session of our work, I would like to thank again the Deputy Chief of Staff, all the officers who have joined us, thank you for your availability, thank you. And of course, thanks to the President of Copasir, to the Deputy Director of the National Cybersecurity Agency, Dr. Ciardi, to Colonel Menegazzo, to all the speakers from the previous panel, and to Dr. Carioti who has expertly managed this difficult round table.

Let us move on to the last session, Information in the Digital Era: True, False, or Virtual? A topic of great relevance and enormous interest. We will have as moderator Dr. Baldo Meo, our press office head for a number of years, as well as an important historical memory of our Authority; Dr. Federico Ferrazza, Director of Wired Italia, whom I invite to take a seat; Dr. Raffaele Barberio, Director of Key4Biz; Dr. David Puente, Deputy Director of Open, television author and fact-checker; and Professor Massimiliano Panarari, communication sociologist at the Mercatorum University and editorialist.

Thank you, the floor is yours.

BALDO MEO

Good evening, I sincerely thank all the attendees who have stoically remained until the end to follow all the sessions. We will reward them, we will try to keep their attention alive with the topics we will address in this session, which indeed concern information in the digital era. Gianni Riotta was supposed to be present, but unfortunately, he is stuck in Lviv.

And we therefore thank Martina Pennisi, journalist of "Corriere della Sera," whom we can also rely on in this panel, in the role that, in my opinion, suits her best: that of a great connoisseur of the world of...

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science, technology, and the impacts that science and technology have on social and economic life. I will now proceed to introduce all the speakers and then I will say something myself, without indulging too much and stealing time.

David Puente is knowledgeable about multimedia technologies, but above all, he is an expert who plays a commendable role in the world of digital information, because he is - we must use the English terms - a debunker, a fact-checker, an expert who exposes "fake news," who counters misinformation and online fraud. Since 2018, David Puente has been the deputy director of Open, where he also holds the position responsible for the fact-checking project.

Next is Raffaele Barberio, founder and editor-in-chief of a well-known magazine in the field of electronic information technology, Key4Biz, and recently the founder of "Privacy Italia," an association very active in promoting data protection.

We also have with us Massimiliano Panarari, a communication sociologist, associate professor at the Mercatorum University of Rome, but also a lecturer in political communication at Luiss and in the history of journalism at Bocconi, as well as an editorialist for important publications such as "La Stampa" and "L'Espresso." He is the author of numerous essays, among which I would like to mention "Poteri e informazione" from 2017, a text related to the themes we will discuss today.

We have with us Federico Ferrazza, director of Wired, also a great connoisseur of tech matters, who also works as a strategic communication consultant. "Wired," it goes without saying, is a prestigious magazine that deals with technology and new media.

With our speakers, we will reflect together on a topic that can rightly be considered crucial and decisive in the era we are living in. In summary, it is the actual capacity of the world of information to build and

form public opinion in coherence with the facts.

And, I add, to do so without manipulating consciences: this is a formula that may seem a bit outdated, but is still useful for understanding the process of transformation and the construction of consensus that can be realized through new media.

We find ourselves in an era that has been defined as the "Platform Society." Well, the platform society, governed by large oligopolists, by Big Tech, is a dimension in which the boundaries between true, false, and virtual have weakened, if not been completely erased. Where for virtual we should begin to understand a pseudo-truth, an elaboration made of a mix of content and sources, tailored for certain categories of users or for broader audiences, with varying degrees of intentionality. And, therefore, also for manipulative purposes.

The systems of this type are known to us: from profiling for electoral purposes in the "Cambridge Analytica" case to fake news and deep fakes, as mentioned by Fausto Carioti and Dr. Ciardi.

To describe deep fakes, the most striking case, perhaps because it is one of the first, involves the Speaker of the U.S. House, Nancy Pelosi, artfully made to appear drunk. Or the video of Zelensky urging Ukrainians to surrender.

In this session, we will also address the protection of personal data in a society of generalized archiving, where the right to be forgotten is something very important, but also very difficult to achieve. A society - according to the reading of philosopher Maurizio Ferraris - of total mobilization.

A world in which we all work for free for the large data monopolists who use it for profit, but also for cognitive purposes, for orientation and influence: every time - and this should now be clear to us - we tweet, comment, or post our opinion, we provide data, revealing our tastes, our habits, our choices, our emotions: all information that will be used to profile us or for predictive purposes. We work for free for the data industry; we are a sort of new species of underclass, so to speak, because we are also depleted of ourselves.

With our speakers, we will therefore try to discuss how our user data - and this happens especially on social media - are used to create informational mass, or rather, a mass of informational pressure, constructed to reinforce or counter an idea, a piece of news, or a pseudo-fact, to amplify it and to build consensus. From identifying tools, our data are becoming tools for creating consensus. We will also inevitably talk about how governments have structured their communication in times of pandemic and war, and how websites and sources have organized themselves for these purposes.

Thus, keeping the red thread of the previous session dedicated to cyberwar, I would like to start by asking David Puente what effects the war has had on information in general and on digital information in particular.

DAVID PUENTE

Good evening.

It is a very interesting topic.

From my experience, I have noticed that there has been this sort of parallelism or fusion between two themes, that of the pandemic and that of the Russian invasion of Ukraine.

I have often heard about these factories of fake news coming from Russia, from St. Petersburg, that...

"they do things," but in the end, I see many small realities in Italy or generally in Europe, of small groups that spread disinformation at the local level. Some do it for money, others for ideology.

Straddling these two situations, I noticed that there has been a sort of alliance based on a concept: "the enemy of my enemy is my friend."

During the pandemic, various disinformation entities created, disseminated, and exploited false news about Covid-19, vaccines, against pharmaceutical companies, against the United States, NATO, or against Bill Gates, accused of wanting to exterminate or control the world. But what happened during the war in Ukraine? The Russians essentially reused a machine that had already been working well during these two pandemic years.

They exploited this environment to propose their narratives against the "common enemies" such as Europe, NATO, and so on. There is a close relationship between the two realities, and regarding this topic, there is an impressive amount of data.

I give an example: there is much talk of fake accounts, identity theft, but I have not found situations of

domains registered with stolen identities; rather, there are situations where personal documents have been modified. We are talking about stolen documents that are used to spread falsehoods, as happened against Zelensky.

If at the European level, not just the Italian one, there is a leak concerning documents, these can be modified to attribute to Zelensky Russian nationality instead of Ukrainian, accusing him of being an illegitimate President. One of the fakes spread against him was precisely this. Certain things circulated freely in the No Vax environment or among Covid-19 conspiracy theorists; that's why they are fertile ground for Russian propaganda.

Taking a step back, I found the theme of privacy and the protection of personal data particularly relevant during the pandemic period. There was a sort of trust between people and the various small entities I mentioned earlier. For many, this trust no longer existed towards institutions, but rather was directed towards a character who might be anonymous or who somehow makes them believe that they will "defend them against the bad guys."

I encountered realities formed by lawyers, doctors, dentists, and nurses who, under the pretext of "treating Covid with alternative therapies," perhaps based on supplements or similar things, ensured that citizens provided their consent to become subjects of study.

I reviewed leaks of documents in Excel format, where there were names and surnames of real people indicating their vaccination status, information about their positivity to Sars-CoV-2, what symptoms they had during the illness, and which medications were provided by which doctors. In some cases, those who treated these patients were not even authorized to do so.

I asked questions to those who conducted a study on these patients, particularly about how consent for the use of sensitive data was presented, obtaining this result: they did not want to answer me.

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How were these data used?

How potentially blackmailable are these people?

In addition to having provided data to this entity, I noticed an excess of trust towards these realities, where a group of doctors promises miracles against the disease but to have priority over other patients, you must register with their committee. Once registered, what happens to the personal data? In this way, a group can boast of having 10,000 or 30,000 members, attributing strength and authority in front of the public, useful for possibly founding a party.

There is too much carelessness, driven in this case by desperation, in providing data to such entities. At the same time, however, the same people demonstrate and denounce the fear that their data will be provided to other realities that may be more structured and more controlled, whether private or the institutions themselves. Among these people, there are those who are afraid to use their health card because they are convinced that "they track everything." I find it absurd that through lies and manipulations, some entities manage to obtain control of sensitive data that they can further exploit to their advantage, not only from a medical point of view but also potentially from an ideological one.

BALDO MEO

If I understood correctly, then, let's say the "hybrid machine" used for the pandemic has then transferred to other areas...

DAVID PUENTE

It's not that it has transferred; it has been reused.

There is something that works, that has worked for two years, that has not spread hoaxes against Russia or Russian vaccines. We are talking about environments that associated problems with vaccines like AstraZeneca or Pfizer, but what about Sputnik?

I was able to read and recount the contents of some clinical records that report adverse events associated with the Russian vaccine, realities that

have not been publicly reported in Russia. They have been shared through the dissemination of documents abroad by a Whistleblower who wanted to reveal what was happening in Russia. The No vax and conspiracy theorist circles, used to spread news about the Russian invasion of Ukraine, do not

talk about it. I am within these social groups to understand how they operate, how they communicate, what they tell, and it is somewhat like entering the rabbit hole, a frightening thing, where they do not contest Russia but rather America, Bill Gates, Soros, and company, always the same ones. So for the Russians, it was, how to say, a red carpet: "the enemy of my enemy is my friend."

****BALDO MEO****

On one hand, the fact-checker has a very important role, but such a role is quite difficult to perform because this kind of fragmentation of sources you referred to is complicated to follow and counteract...

****DAVID PUENTE****

There are many realities, and it is not just an Italian problem. During the pandemic, with my foreign colleagues, about seventy of us around the world, we agreed, communicated, and exchanged information about what was happening, noting that a hoax spread by a small group in Paris or France could become viral even in the United States or South America. Many small realities that make noise and then suddenly explode. We are required to understand what is happening and to somehow monitor all these networks; you should see one of my phones where I keep all the Telegram chats I monitor, a very long list of channels, and I struggle to manage the notifications. I conduct research through keywords, especially when I am alerted to a post on social media, and I discover where it was shared from and in what manner, possibly finding the origin in a Russian Telegram channel or a French one.

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One can reconstruct a path that has been taken from the "news," and it is difficult because we are faced with a mosaic made up of small pieces, in a somewhat frightening sense because you do not know what they might invent among the various realities. There are many minds; there is not just one.

****BALDO MEO****

In this discourse, there is a background presence: social media. And I ask Raffaele Barberio: if the fact-checker deconstructs false news, who constructs it?

****RAFFAELE BARBERIO****

Reality and its distortions that aim to undermine the protection of personal data. Social media do not present objective facts; they offer, without apparent control, what users cut out and publish on this or that channel. We can say that there is no more or less relevant intervention from that sophisticated artificial intelligence recognized in social channels. What appears instead is the duality between reality and representation, much older than the short life of the internet compared to the ways in which we have had to face this issue over time. How many times have we seen a freshly cut rose in the garden, so perfect that it makes us exclaim: "How beautiful, it looks fake." And how many times have we observed a plastic or fabric rose, like those seen in some sophisticated offices, made so well that it makes us comment: "How beautiful! It looks real." It is a metaphor to say that things are never as they appear. And wanting to remain on the ground of metaphors, we are often dramatically revealed with a new course of events or the occurrence of circumstances in open contradiction to the rules of common sense.

Even in this case, there is a sense of the search for novelty at all costs. But it is not so. The world has always turned the same way, and perhaps for this reason, the first rule is not to trust what you see. Things are almost never as they appear.

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Now let us try to apply this point of view to the topic of the moment, namely the current state of the war in Ukraine and the ways in which the reality of the facts is represented by both sides in the conflict. It is difficult to imagine that on one side there are all liars and on the other there are all truth-tellers. War alters everything, not only the battered territories or the ordinary lives of people, which are completely

disrupted. How to face and manage information during a war? From the inside, one of the fundamental rules is to discredit the enemy. Again, there is nothing new here. Yet, in these weeks, there has been continuous talk of misinformation, of the Russian machine of lies. These are statements that risk, in turn, contributing to altering reality and hide the fact that the objective of discrediting the enemy in the eyes of national and, even more, international public opinion has always been one of the fundamental practices of a state of war. On both sides involved in the conflict, there is nothing but the interpretation of a script as old as the world.

The methods of information

Who does not know, after all,
international relations during the Cold War?

Who does not remember the saga of Radio Free Europe, the CIA's radio station set up in Germany, or The Voice of America, also in Germany on one side, and Radio Moscow, on the other, broadcasting in 74 languages?

Then, as now, the objective was to discredit the enemy, with a mutual disinformation campaign that is an integral part of a state of war and the use of all the weapons it implies.

But let us return to the war in Ukraine; we continuously read about new forms of so-called "hybrid warfare," but in fact, every war has always been hybrid because it is fought with every type of weapon, both real and virtual, cultural and military.

So when I read or hear from newspapers and TV about this great novelty of hybrid warfare, I cannot help but smile.

What is real, however, is another hybrid war, but undeclared,
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that is fought every day on the internet, with the extraction of data from citizens of this or that nation, aimed at acquiring data on which commercial dominance can first be exercised and perhaps later political or even mass surveillance.

This is a data collection carried out in a context of apparent consent, because the mountains of data that transit on the internet to be delivered to so-called third parties are voluntarily provided with no small amount of superficiality by the users themselves. A singular Stockholm syndrome in which the victim does not realize what they are feeding to their executioner, ignoring the very nature of the latter. From this point of view, social media is the greatest war machine for the collection of personal data. There has never been such a belligerent machine in the history of humanity, and all this has happened in just twenty years or so. What are these data collected for? Certainly, the answer seems quite obvious, but once again, things are not as they appear. They tell us that their collection is useful for better interpreting personal tastes and offering a better quality of service. But it is not so, because even this goal is not an end but a means to achieve something else.

So are they collected for the purpose of trading data? Here we begin to hit the mark. We know that Facebook was not created to facilitate communication among people, but to collect data. We know that Google is not designed to provide us with a search engine, but to collect data.

We know that Microsoft allows us to send and receive emails, but it does not seem to matter that it hoards billions of data contained in our emails and uses them in the most disparate ways. The same consideration applies to the Chinese TikTok or to Amazon's commerce. Then there are purely IT players, like Oracle, which have perfected the millimetric extraction methods of data with techniques that until recently were inconceivable. It is worth noting that Oracle's software is at the core of mass surveillance systems in China. An alarming phenomenon that does not only concern China or so-called non-democratic regimes, since the city, after Beijing, with the most penetrating mass surveillance system is London, the home of the first democratic Parliament in history.

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Thus, the substantial business model of these companies is primarily to collect data. At this point, one might ask: what are these data for and why are they collected in such an intensive manner? The first answer, supported by commercial motivations, is that their collection allows many companies to develop their business in a more sophisticated and targeted way, in strongly personalized forms.

There are hundreds and hundreds of companies in the world that collect data through social media according to filters agreed upon with the social media themselves. These data, once collected, become raw or processed goods for hundreds of companies that use the personal data collected from hundreds of millions of people. They are the so-called personal data brokers, who collect, buy, process, and sell personal data wholesale and retail, using all the tricks that the internet allows. Let us look at a few of them. Axiom, for example, has a database of one billion people worldwide, of whom it knows all real estate transactions. Another, Corlogic, much smaller, does more or less the same work. There is yet another, Equifax, which perhaps all of you will remember because it was the subject of a hacker attack that stole the data of some hundreds of millions of people. In fact, Equifax has a long history of data collection, having been established as far back as the late 1800s. Finally, Nielsen itself has always collected data and has also been around for over a hundred years.

It should be noted that, even in these cases, there is nothing new; what changes is the way in which data collection methods are made efficient. And here the extraordinary capabilities of digital technology emerge.

But I would like to return to my initial invitation to consider that things are never as they appear. The initial driving force of mass innovation, starting in the 1980s, relied on a compelling narrative that was as fascinating as it was captivating, recounting the intriguing stories of young tech enthusiasts, often lacking any formal education, holed up in garages of American homes, capable of building visions and inventing unprecedented technological solutions.

But this is the narrative. Certainly, those young people were flesh and blood and had talent to spare, but the origins of the technological miracle did not lie in their heads.

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The technology, all or almost all of the digital technology we use today, is a direct offspring of the immense military research and development effort of the 1950s, which in turn was generated by the Cold War between the USA and the Soviet Union.

The clash between the two superpowers was fueled by a competition over the availability of new weapons, and within this process, all technological expressions of the era can be linked, from the race to space (always motivated by military reasons) to the production of chips capable of ensuring ever-increasing computing power. Silicon Valley itself, born in the early 1970s, can only be considered as the industrial expression of that movement, and Reagan's subsequent Strategic Defense Initiative, which marked the beginning of the military collapse of the then Soviet Union, confirmed how the power of a superpower could be exercised primarily thanks to the available military technology.

Naturally, the military research and development effort must then be converted for civilian uses. It is an inevitable mechanism that allows for a renewed impetus to the industrial structure, economic growth, and the development of every nation.

Looking through the case of the USA, the processes of growth of the American economy in relation to the technological commitment of military apparatuses is emblematic. For example, if we look at today's Mediterranean, the country with the greatest technological capacity and with an army (one might say) of startups capable of finding practical solutions in every sector of daily life is the State of Israel, which, not coincidentally, has been in an uninterrupted state of war since the nation's founding in the late 1940s.

But let's return to the USA to see what has happened more precisely. What has happened is that for decades the so-called Military Industrial Complex (MIC) has exerted a decisive weight in the American economy, steering the economy and international relations within a broader geopolitical framework.

For the MIC, conducting military research and development was a must. And the technological acquisitions coming from the military sector have fed a pool of knowledge that has begun to generate

those innovations directly oriented towards civilian use that are evident to all. Then, at a certain point, American civilian technology, namely Silicon Valley, began to run faster, but much faster than the pace of the research and development apparatus of the MIC, to the point that those in Silicon Valley began to tell the military: "...That project of yours requires twenty months of processing, but if you give it to us, we can complete it in a quarter of the processing time...", and the roles were reversed. Thus, military research and development fed a technology that began to reproduce and grow autonomously to the point of becoming a support for the military, eventually taking control of operations.

Today, it is not so easy to distinguish the two segments; technology is unique and is distinguished only by families of applications in one or the other field. And, as is known, there are decision-making dashboards composed of representatives from the military and representatives from Big Tech. One of these dashboards sees American generals sitting at the same table as figures like Eric Schmidt or Jeff Bezos.

In light of these considerations, the war in Ukraine is situated in a context that is less flat and less obvious than...

as far as it appears.

The data, in this war conflict, fall among the most used projectiles.

The clash in the cyber domain is as important as infantry attacks or breakthroughs with armored vehicles.

At the same time, the war front is far from being limited to the border line between Ukraine and Russia. At stake are much more complex balances: the composition of NATO, the relations between the EU and the USA, the dynamics among the countries within the EU, the autonomy of the latter with respect to the unipolar aspirations of the USA.

On April 18, 2022, a letter signed by Leon Panetta (former Director of the CIA), Michael Rogers (former Director of the National Security Agency), Frances Townsend (former Assistant to the President for Homeland Security and Counterterrorism), Michael Morell (former Deputy Director of the CIA), James Clapper (former Director of National Intelligence), Jane Harman (former member of the House Intelligence Committee), and finally Jeh C. Johnson (former Secretary of Homeland Security) was circulated in the USA, within institutional circuits, although later leaked to the outside.

Seven distinguished signatures, all retired individuals.

But in American culture, when you have served in certain offices, even if you are retired, you remain a patriot and you are always a soldier.

Why this letter?

These gentlemen stated a very simple thing: there is a shared fear among most members of Congress and the American institutional world that the European Digital Market Act may only benefit powerful state-owned enterprises or Russian and Chinese subsidiaries, and therefore they request that Europe amend the Digital Market Act. And in the body of the letter, explicit references were made to the effort that American platforms are making to counter the enemy that threatens the borders, the economy, and culture itself.

Now, seven retirees saying these things, it could be said, do no harm to anyone, but the principle is that they speak to the upper echelons of American politics, and what they express is a strong pressure on Europe to amend a law that was in the process of approval at that time.

In this way, by combining conventional weapons and network platforms used for entertainment, one is merely "weaponizing" technology. All of it. And the same platforms are, in fact, being "weaponized" to be used as a cog in the war mechanism.

Essentially, the meaning of that open letter to Europe was: Europeans, do not slow down the development and action of our platforms, because our platforms are necessary to neutralize Russian disinformation.

The point is that everyone engages in disinformation, and therefore, it is not clear why to use the dramatic occasion of this war, in which Europe is not directly involved, to legitimize operations that undermine the European regulatory system.

This war has somehow diminished European autonomy, the autonomy of a continent made up of 500 million citizens who have their personal data. If Europe loses its autonomy or must subject its

prerogatives to the needs of other nations, what happens to the data of these 500 million citizens? This is the political problem that we all must pose to ourselves.

Fortunately, the European regulatory framework is holding up, but there remains an ongoing clash that is far from being quelled.

We face a world that is perhaps destined for regional fragmentation due to the pressures towards deglobalization. Winds of war may continue to blow for many months, but regardless of this state of conflict, the defense of the integrity of the personal data of European citizens must be safeguarded, because their protection is at the foundation of some fundamental principles of our democracy.

BALDO MEO

Thank you, Raffaele Barberio: you have put a lot on the table. I take the suggestion of the "weaponization" of technology, which seems to me an issue that requires reflection, this entry into the war machinery by technology. Personally, I am not so convinced that history always repeats itself because a hybrid war machine as powerful as social media has never existed in history.

I now give the floor to Massimiliano Panarari and ask him what his vision is, as a sociologist of communication and information, of the current situation. Today, is traditional, or rather professional, information truly so devastated by social media, or does it have its own validity independent of certain tools?

MASSIMILIANO PANARARI

To paraphrase General de Gaulle, one could say that it is a "vast question."

Good evening everyone, and thank you very much for the invitation to the Authority.

I believe there are a number of plans to address in relation to what was raised by Baldo Meo, and also with respect to the previous interventions that have been very rich in insights, among them very different naturally.

I believe there is a directly industrial level concerning information, or a plan - as it would have been said in another era, and as an elderly gentleman with a beard would have said - structural, from which one cannot prescind, because information is - as is known - a commodity, therefore it must be sold just like any other type of product, and on the other hand, it is a fundamental component of the processes of constructing the public sphere. Commodity, with reference to the first dimension, which has been (and is) sold in various ways; and this is why the Yellow Press and the sensationalist press of late 19th century New York claimed, for example, that fossils of a dragon or a unicorn had been found at Coney Island, naturally to increase the number of copies sold.

At the same time, precisely, in the fatigued Western society, information also presents and even more that component that could be defined in some ways as value-based, which was evoked this morning by the President and the other members of the Authority: that is, essentially starting from the dissemination of Enlightenment ideas, it identifies a pillar upon which the public sphere has been built. The crux - as is known - is that today information in itself is not sufficient to structure, empower, and give authority to the public sphere. And, moreover, it is constantly under attack from multiple subjects, starting with hostile powers that resort to the tools of hybrid warfare and cyberwar.

But, more generally, also from other actors, such as economic powers that today return with great frequency within the discussions held on this day, and determine a push towards the privatization of public space and the orientation of public discourse, from which derives a weakening of the public sphere.

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An additional element concerns the users' side, in the sense that consciously or unconsciously the transformation of information is heading towards a horizontalization and total participation of each of us in what is no longer precisely the public sphere in the classical sense of modernity, but coincides, in the best of cases, with an "interrelated public sphere," to which extraordinarily different subjects participate compared to those of its Enlightenment genesis and which arrives "intact" in such form until the First World War.

It is precisely the conflict of 1914-'18 that projects the populations involved into a dimension of

unprecedented technicalization and scientificization of propaganda compared to the past, which will subsequently lead different subjects, no longer exclusively state and public powers, to compete for the conquest of a consensus that is no longer shared and is fragmenting, and this constitutes the further, profound difference.

If we observe what has traditionally been called the public discourse, which was essentially governed, structured, and homogenized by information operators (starting from the gazettes of the French Revolution, the American Revolution, and the great liberal revolutions), we find that that shared plane has naturally collapsed and the language of public discourse is no longer homogeneous, while everyone competes within what should be a place of sharing, of dialogic formation, the famous public sphere in the Habermasian definition.

It is, we could say in a literal sense, a condition of competition, primarily in the field of the attention economy, one of the major underlying knots that have undermined the shared strength and the ability to construct a shared public discourse by information subjects. An attention economy in which competition we find the great paper press, but also and above all today a very strong push from TikTok, to give an example.

And so if we read a news item in a paper newspaper or even on the homepage of what is called a mainstream newspaper for

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example regarding drought, we can compare it to - here I am somewhat oversimplifying to try to be explanatory

of course - we can

compare it to the ballet of a trained cat

on TikTok? Clearly, the answer is no. But, precisely,

all these subjects and

media actors compete equivalently

("horizontalized") in the

market of the attention economy. Because - and

this is another significant aspect

to consider - what has disintegrated is,

clearly, the dimension of authority.

This is not a new problem. To find a symbolic date

- very

debatable of course like all those adopted

to evoke turning points -

one can refer to the crisis of authority

after the Sixties, an effect

of the processes of hyper-subjectivization and

hyper-individualization that reach,

naturally relaunched with even greater force,

up to our times.

This process has made it increasingly difficult

to find a

shared cultural and symbolic space to which everyone

attributes

the same value of authority. Here, if we can

say that within

this magnificent hall in Turin where the conference is

taking place, it is

plausible that the people present identify as

authoritative sources among

themselves very similar (if not directly all the

same).

The problem, however, is that “out there” this sharing of the idea of authority no longer exists, but without it it proves extremely difficult, if not decidedly impossible, to collectively inhabit a public space. And, indeed, this can be unfortunately measured at every level of the immaterial infrastructures of our society.

Consider the public school and the disintegration it has been undergoing for some time, a process fueled by social issues as is well known, with an exacerbation of inequalities that produces a return to illiteracy.

There is now a clear difficulty in having truly shared cultural tools, because what has come under crisis is precisely the idea of a public sphere and the recognition of authority along with the traditional mode of transmission of knowledge.

And, therefore, a clear awareness is needed of the fact that

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that undeniable and significant work of democratization linked to the process of horizontalization of the media highlights, also, of course, a very problematic dimension (and, indeed, “poisoned”). If we all participate in the public space not on the basis of a shared educational background, not on the basis of acceptance of the same “rules of engagement,” not on the basis of recognition of expert systems, the already fragmented public sphere moves towards decomposition and a sort of total drift.

Certainly, in addition to these considerations, we cannot hide the fact that even some specialists bear their share of responsibility, and this has been seen in the public discourse on digital media during the pandemic when we were deluded for a significant initial phase that, in the tragic context, it

was also a window of opportunity for the public opinion to reassign credibility to virologists, infectious disease specialists, immunologists, and, more generally, to representatives of scientific knowledge. But we also had to witness their rapid subjugation to media logic, also due to a series of very pronounced individual psychological dynamics (the search for publicity and personal visibility), which has led experts to “horizontalize” and yield to a series of criticisms. Here we are somewhat oversimplifying, but if an authoritative infectious disease specialist agrees to be interviewed about his views on love and marital fidelity, and several of them tell us in a glossy magazine how they will spend their August vacations at a time when, still until not too long ago, they were saying that one should essentially not go on vacation to avoid the return of contagion, then we witness a settling into and acquiescing to the mechanisms of media logic, placing themselves indeed at the level of anyone else (a sort of paradoxical “one is worth the other”). All this happens also because the transfer of the driving force of the media, which compete for the attention economy, onto a plane of spectacularization, is very strong.

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It is what a German-Korean philosopher—sometimes a bit overrated but very much in vogue and often interesting—Byung-Chul Han, calls the society of entertainment. In the sense that the only authentically universal (i.e., global) code that remains today coincides in fact with entertainment: currently, this is truly the basis of sharing, substitutive of authority; and, clearly, this process produces a different context, and, indeed, we could say a sort of storm that descends upon our heads and our public debate.

What role do platforms play in this dynamic?

A very strong, decisive role.

Our platform society, the platformization of public debate, which horizontalizes and disintermediates everything, delivers the scenario to a dimension of equivalence of opinions; it is the famous debate between doxa and episteme, with the prevalence of doxa and a deconstruction of trust and credibility. Credibility and trust are two elements that are in a dynamic relationship, a relationship that changes over time, and shifts with the material and immaterial infrastructures of the social body.

Ours is a society—David Puente said it—in which trust is declining, and therefore, conversely,

credibility towards institutions is also diminishing. Those institutions, as has already been mentioned during this day of work, are strongly associated with public debate and information and are the pillars of liberal-representative democracy.

What is growing, as a counterbalance, is not a form of general and shared credibility, but rather affective polarization towards those whom David Puente always referred to as small friendly groups, small communities, spheres of peers in which the fundamental component of distrust towards those considered other than oneself comes into play, because they are positioned from the perspective of social hierarchy at a higher level.

And here, then, everything that can be traced back to elites and establishment is separated from being a bearer of expert knowledge and is considered akin to a bearer of a form of social privilege that is odious and incomprehensible.

The public sphere needs a rational dimension, as is evident, especially at a time when the sentimental and affective component and, above all, that of sad passions, as Spinoza called them and more recently Miguel Benasayag, becomes the only true social propellant, with the risk of potential implosions.

Today, in the best-case scenario, we live in a context of an "interrelated public sphere," in which qualitatively different subjects participate, and which therefore needs to be defended as a dam and as a barrier—and the protection of personal data represents one of the most fundamental of such barriers.

But we are also increasingly moving towards a multiplication of public spheres, which in some ways brings us back to a typical stage of pre-modernity. Before the Enlightenment, there was no unitary public sphere; there were many public spheres, small, extremely limited, non-communicating with each other, characterized by deep social inequalities.

If the postmodernity into which we have long since plunged represents, under various profiles, the re-emergence and re-proposition of a series of dynamics characteristic of the pre-modern, today is precisely one of those. With the fundamental difference that back then there were no platforms, whereas today we must deal massively with them.

BALDO MEO

Thank you. Crisis of credibility, crisis of trust, lack of a public discourse...

At this point, I would like to reflect with Federico Ferrazza on a specific topic: true, false, and virtual.

Are there news items made solely of perception?

In other words, can we understand the concept of "virtual" as pure perception?

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FEDERICO FERRAZZA

We are focusing a lot on social networks, on algorithms, on how we are manipulated and on the attempt at manipulation by the algorithms themselves or by foreign forces. But I invite you to reflect on something that perhaps pairs with what Barberio said a little while ago (namely that there is nothing new): have you seen what television broadcast during the war?

I am referring to the information about the war on TV or in newspapers. As anti-deontological as possible.

And the same goes for Covid. I do not think the problem is only Facebook, Instagram, TikTok, Telegram.

The lies that have come out regarding the effectiveness of the vaccine, the idea that they want to catalog us, the notion that we are at war on behalf of someone or that Ukraine is a nation of Nazis in the pay of Europe or the United States to defeat Putin have appeared in newspapers, in publications registered with the Italian courts. It's not that they appeared on social networks.

So before starting to find blame and responsibility in social networks or algorithms, perhaps I would first look at home.

And I see that there is a lot of misinformation being produced by traditional media outlets that have nothing to do with social...

network. Then I think that disinformation (I don't really like to talk about fake news because

determining what is true or not is always quite difficult) is comparable to a pandemic. And how do you defeat a pandemic? You defeat it by vaccinating people. Therefore, in my opinion, the only way to understand the effectiveness of a disinformation campaign is to understand which subjects it targets, and unfortunately in Italy it targets a rather ignorant citizenry.

We were talking about perceptions: there is a beautiful study by Ipsos that has been ongoing for several years, called "Perils of Perception," which means the risks and dangers of perception. In the study, it shows how Italy is one of the countries, if not the country, whose citizens have the greatest gap between perception and reality. Moreover, according to Istat, in Italy, among those aged 30 to 34, 75% have a diploma, while the EU average is 84%; and in the same age group, 27.6% are graduates, making us the last in Europe along with Bulgaria, where the EU average is over 40%.

So, when we talk about disinformation, there are also those who believe in false news, and the less culture there is in the country, the more people are inclined to believe it. I mentioned the Ipsos research: from 2000 to today, the number of homicides in our country has decreased by 40% each year. 49% of Italians think it has increased; 35% think it is the same; only 8% respond correctly. Italians believe that 30% of those living in Italy are immigrants. The truth is that it is only 7%.

And regarding Muslims, the question posed by Ipsos was: how many Muslims are there in Italy for every hundred inhabitants? The average response was 20, while the true answer is 4. And then: how many people over 65 are there for every hundred inhabitants? The response from Italians is forty-eight, while the correct answer is 21.

The next statistic is a bit older but helps to continue the reasoning: how many unemployed people are currently looking for work in Italy? According to Italians, it is 49%, while the truth is 12%. All of this gives a measure of how poorly we are educated, not only from a digital perspective, and how fertile a ground Italians are for what we call fake news or disinformation.

It is quite evident that the only way to counter all of this is to try to increase knowledge, invest in education, and on the other hand - I am talking about the world of information - much more work should be done on awareness and ethics; I resonate a lot with what Panarari said when he talked about entertainment. Today, the informational offerings of newspapers and television programs that provide news are mostly constructed with the logic of entertainment.

Because when the usual people are called on TV to talk about war, Covid, Draghi, Mattarella, or any other topic, it means that an informational schedule is being constructed that is made like an episode of *Uomini e Donne* or any entertainment or variety program, and I apologize to the authors of *Uomini e Donne*.

So it is pure entertainment. Then there are the responsibilities of social media. More than responsibilities, I observe the structure, which is certainly not designed to help combat disinformation. All platforms are made to polarize discussions. Every video, photo, post, opinion, idea, or any type of content is always put to a vote by a heart, a retweet, a thumbs up, or a comment. But reality is made of nuances and complexities, and excessive polarization does not help the proper unfolding of information.

And then there is an interesting biological issue supporting all of this, which is the issue of dopamine. Dopamine is a neurotransmitter that gives us pleasure every time we receive a token of esteem or another form of approval. Several studies have been conducted showing that people then spend time checking if they have received a like, if they have received a heart, if they have received a positive comment on their Instagram photo because all of this gives dopamine bursts. All of this is to understand what the structural mechanisms of social networks are and how the entire engagement system is constructed, to put it in English, that occurs in those digital spaces.

BALDO MEO

Thanks to Federico Ferrazza, who has given us a refreshing dose of reality: he has brought us back down to earth, where there is friction, and has invited us not to blame everything on social media and, perhaps, not too much on the "emotional contagion" of which they are bearers. I would ask Martina

Pennisi, to conclude, for a word of comfort on the fact that, after all, serious and authoritative information can still be done.

MARTINA PENNISI

I also thought it could be a good way to close. I do so actually by reconnecting to what was said this morning in the session I moderated. In the panel, there was a representative from Meta, formerly Facebook, the 168 LA PROTEZIONE DEI DATI. DA 25 ANNI LA BUSSOLA DEL FUTURO

****The most important social network group in the world, which has over 3 billion users, was here to discuss how they are trying to innovate and experiment with other business models, imagining and building the Metaverse. We, rightly, are still here questioning the effect that social networking platforms have had and continue to have on information. In this sense, information, the work of newspapers and journalists is not very different from that of regulators and legislators in the face of innovation and technology. I'll give a very quick example: those who work at the desk, meaning those who are inside the newsrooms and not outside following events like this, have a whole series of tools and programs where they see real-time data, so they see how people react to content online, because engagement is obviously a data point on user reactions, on how the content of competing publications is performing. Well, this slot machine is also exposed to information professionals. And yet, for the platforms, newspapers have been a thorn in their side, and they still are for issues related to copyright, and a whole other series of reasons related to fact-checking. These are aspects to consider, aspects we are analyzing in light of the fact that the future is a continuous race and we must filter – we professionals in the field – without letting ourselves be too swept away by this continuous availability of data and information, the effects of which have been very well explained in this panel and are quite clear.****

****BALDO MEO****

Thank you, Martina, we needed your words after so many issues. Let's conclude with a final reflection from David Puente and then I'll hand the floor to Dr. Ghiglia for the closing remarks.

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****DAVID PUENTE****

As a final reflection. With Open, I deal with fact-checking within the social network Facebook, so when any of you shares something false, you might find the notification "This is false news" or "partially false" or "lacking context," a news item verified by an independent fact-checker that provides an article explaining all the issues. I see this practice as a sort of preparatory action, where there is no censorship, but a notice and the intention to explain what is happening and what has been disseminated incorrectly. We often criticize social networks for various reasons; no one is perfect, but let's realize that they are just means. Lies have always existed, hoaxes have always existed, propaganda has always existed; the means change, the methods change, and even the studies of what we do in these new realities change, but I wouldn't want to demonize them entirely. Activities like the ones I do can help, but they are not sufficient: fact-checking is not the solution; it is an additional tool to combat this sick system. I say sick to emphasize an exaggeration. However, I would like to leave this discussion by saying that it is somewhat the responsibility of all of us; each of us is responsible for what we share, each of us is responsible for what we tell others, because all of you in this room have credibility with the people who follow you, just like famous influencers or virologists who speak on television, but each of us, in our small way, is responsible. We can also "get angry" at the platforms and certain realities, but if each of us does our small part in the fight against misinformation, as we fact-checkers do, it wouldn't be bad.**

****BALDO MEO****

Thank you, David. This is also what the Guarantor has always recommended: we users need to take responsibility to use the new tools that technology offers us consciously.

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Let's conclude this session by thanking you for your attention, and I'll hand the floor to Dr. Ghiglia.

****AGOSTINO GHIGLIA****

Thank you, thank you all. In thanking the Eventum Society and Dr. Giani for the wonderful organization, I would like to take another two minutes with a contribution, our spot; it was the first spot of this new Guarantor, not to celebrate the twenty-five years but to remind a bit of what our mission has been from the beginning, that is, as we said this morning, to protect people's data, thus protecting our freedom. Thank you all also for your patience, for your attention, for the intelligence with which you have followed this event. And I won't say that we will see each other in twenty-five years; I hope to see many of you much sooner, but there will certainly still be a need, even in twenty-five years, for a renewed Guarantor, for a present Guarantor, for a Guarantor that is a bit of a shield for our personal freedoms.

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